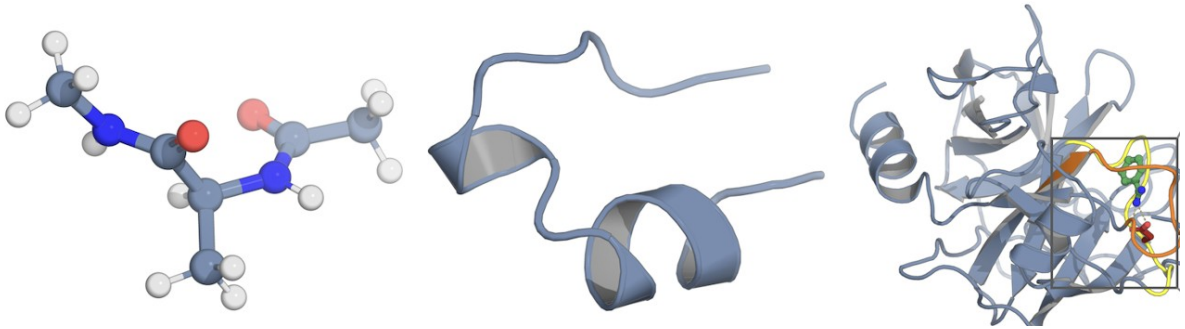
Representation and Reinforcement Learning for Molecular Dynamics

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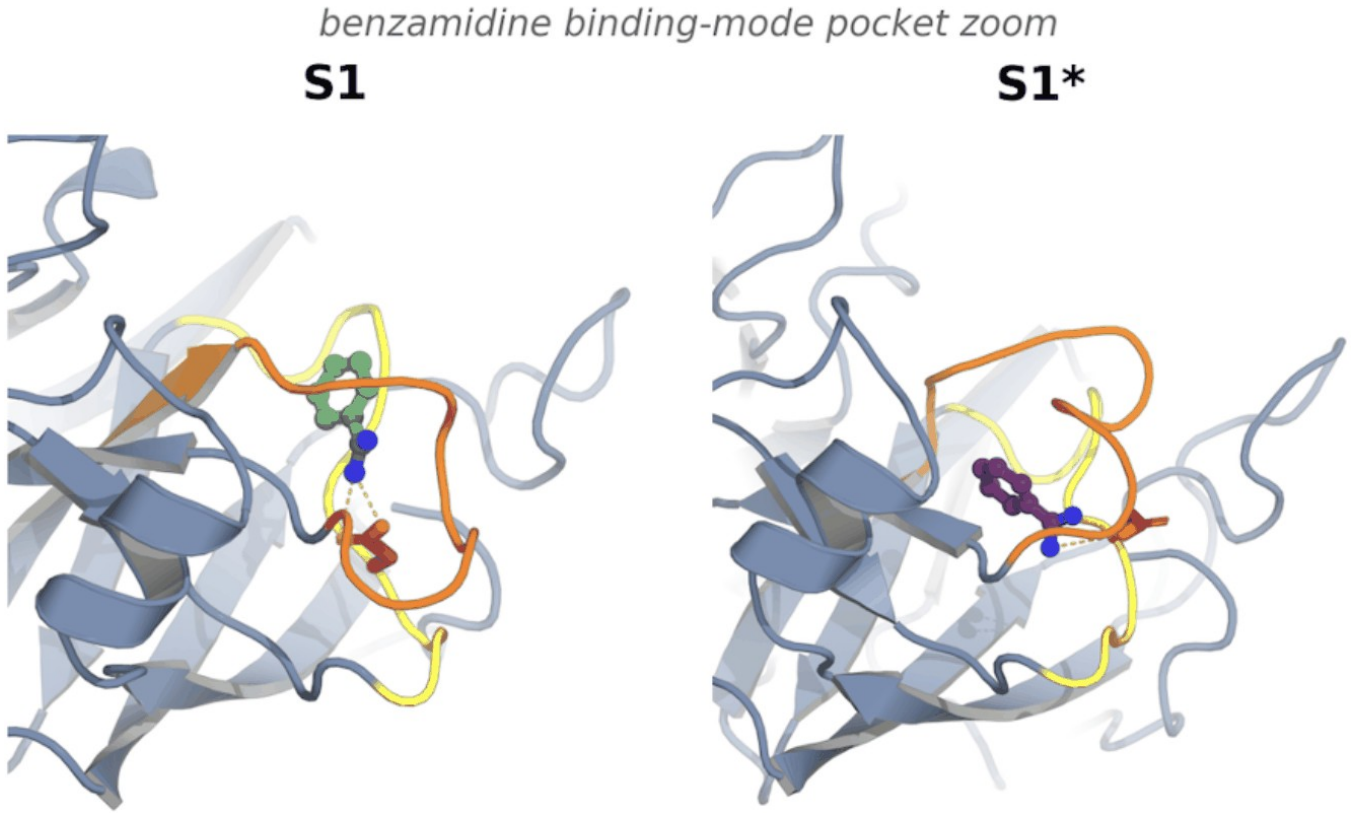
Advisor: Aaron R. Dinner



Alanine dipeptide | Trp-cage | Trypsin–benzamidine

What to represent, and where to simulate, in molecular dynamics

A binding pose does not reveal a binding mechanism



S1 native mode, Trp215 loop closed **S1*** alternative mode, Trp215 loop open

Trp215 loop orange · Asp189 loop yellow · Asp189 side chain red · Asp189–benzamidine salt bridge dashed

Route

How does the ligand enter or leave?

Receptor motion

What changes in the binding pocket?

Ordering

Which change occurs first?

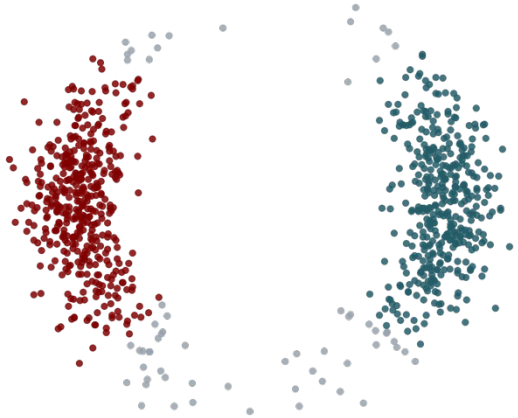
The target is the dynamics connecting these conformations

Trypsin–benzamidine structures; S1/S1* binding modes as defined by Plattner & Noé, Nat. Commun. 6, 2015

Structures show where; MD shows how and how fast

Independent structures

Where the states are

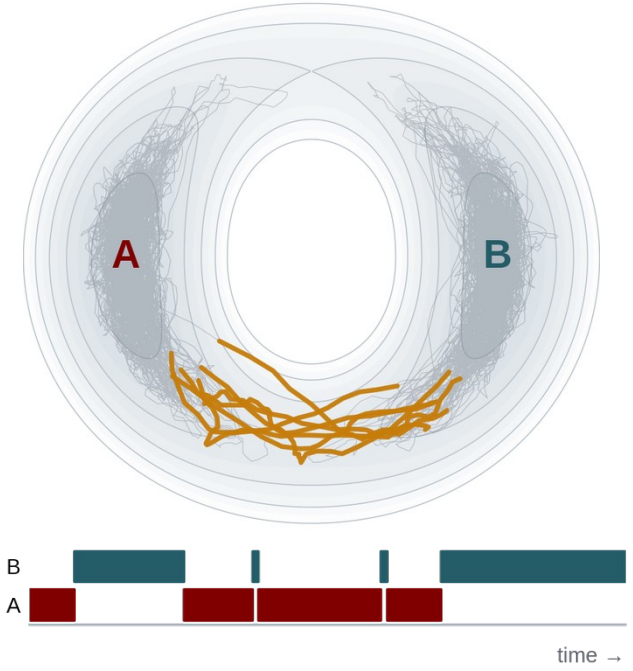


Identical for both landscapes

The two ensembles differ by less than 1%, and few samples fall between the basins.

MD on landscape 1

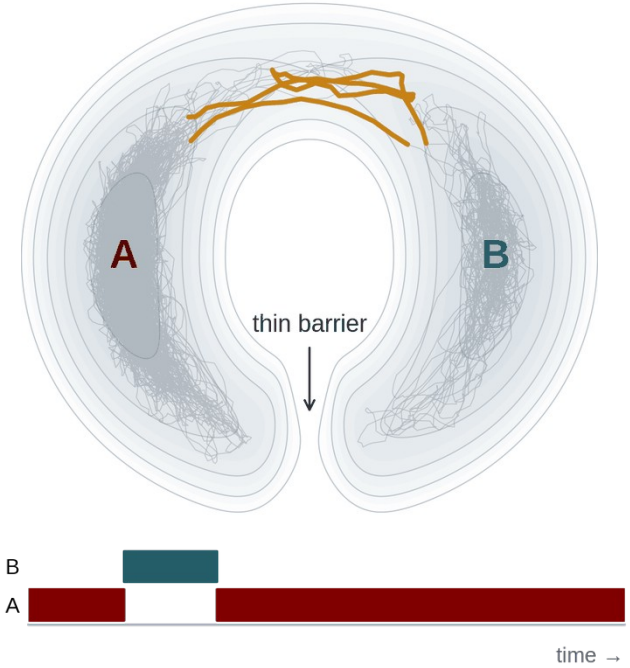
How and how often they connect



Mostly the lower route

MD on landscape 2

With one thin barrier

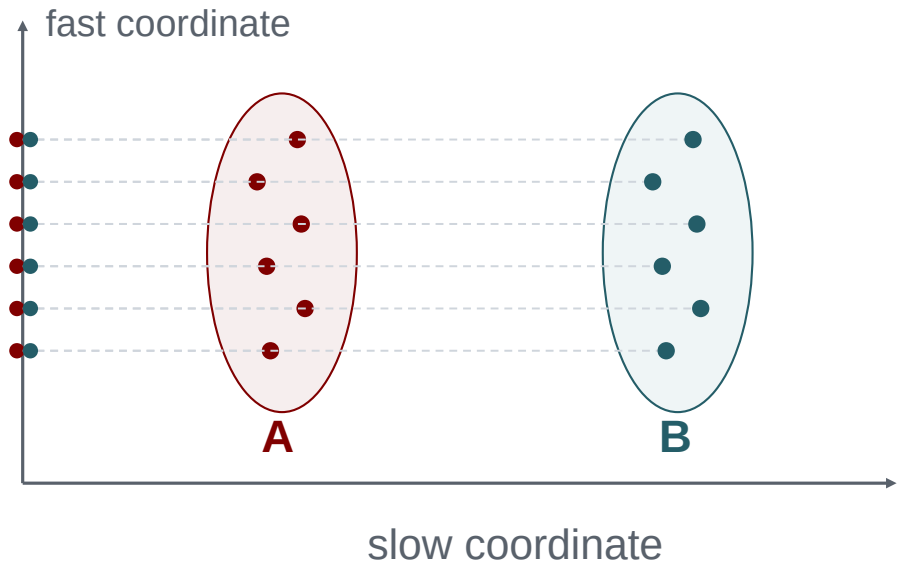


Rerouted over the top, fewer transitions

Toy 2D model, overdamped Langevin dynamics (kT = 1). Ensemble difference and rates from 200 independent trajectories per landscape.

Two bottlenecks of MD

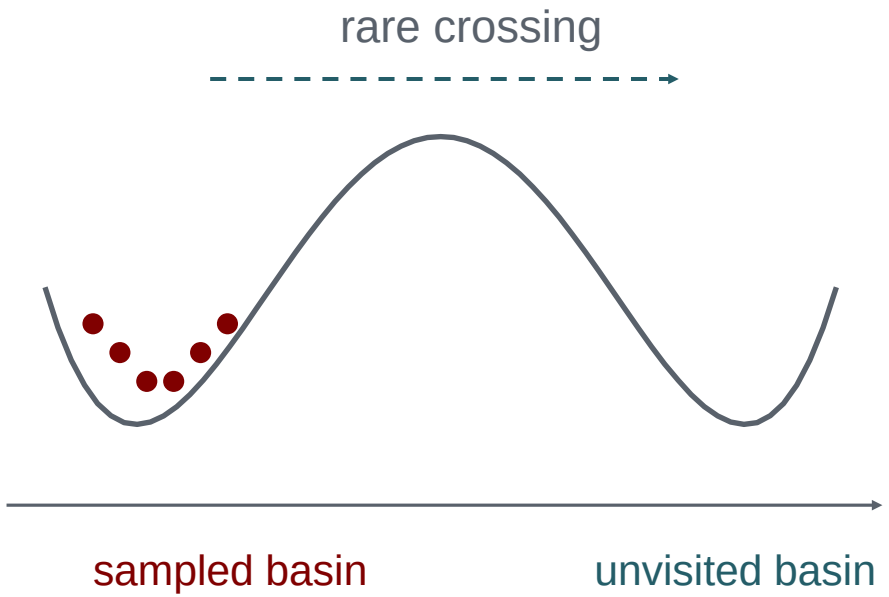
Representation



Both states are present,
but their distinction is lost.

Representation learning: keep slow distinctions visible

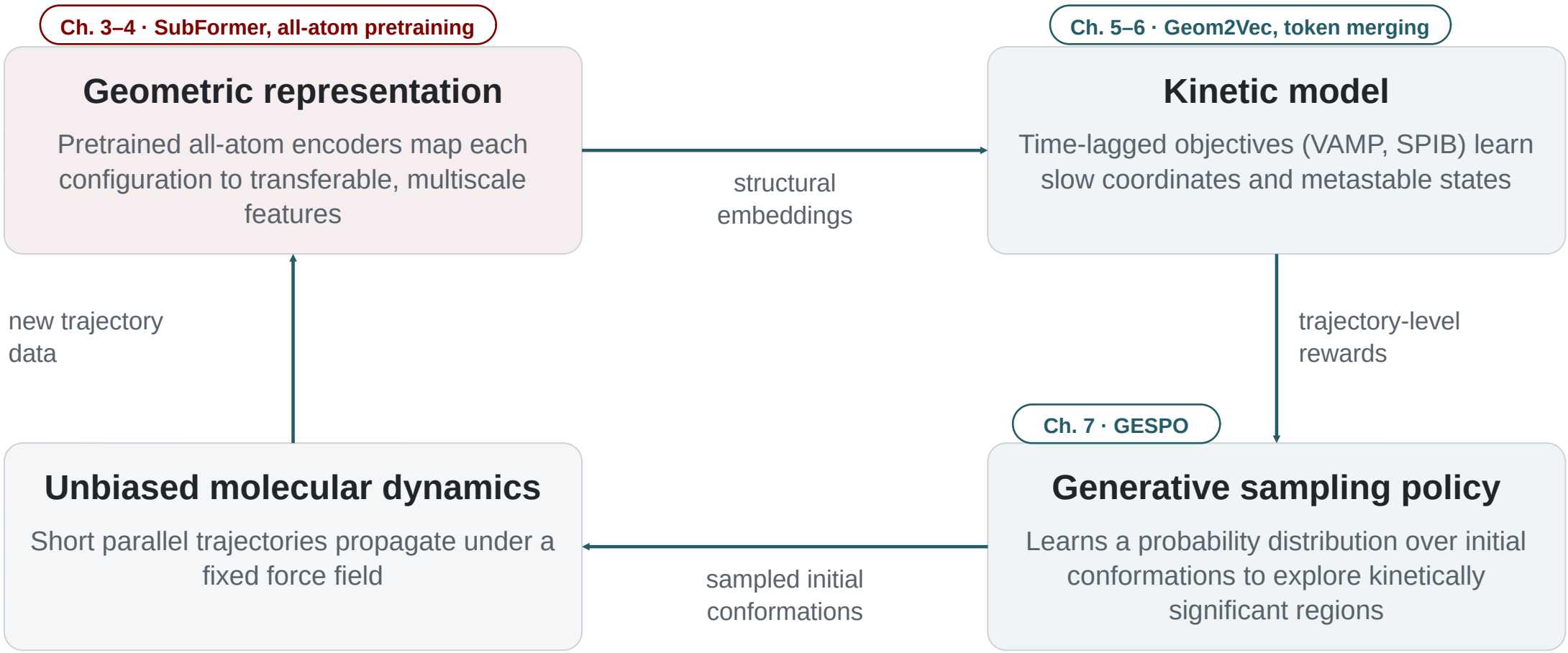
Sampling



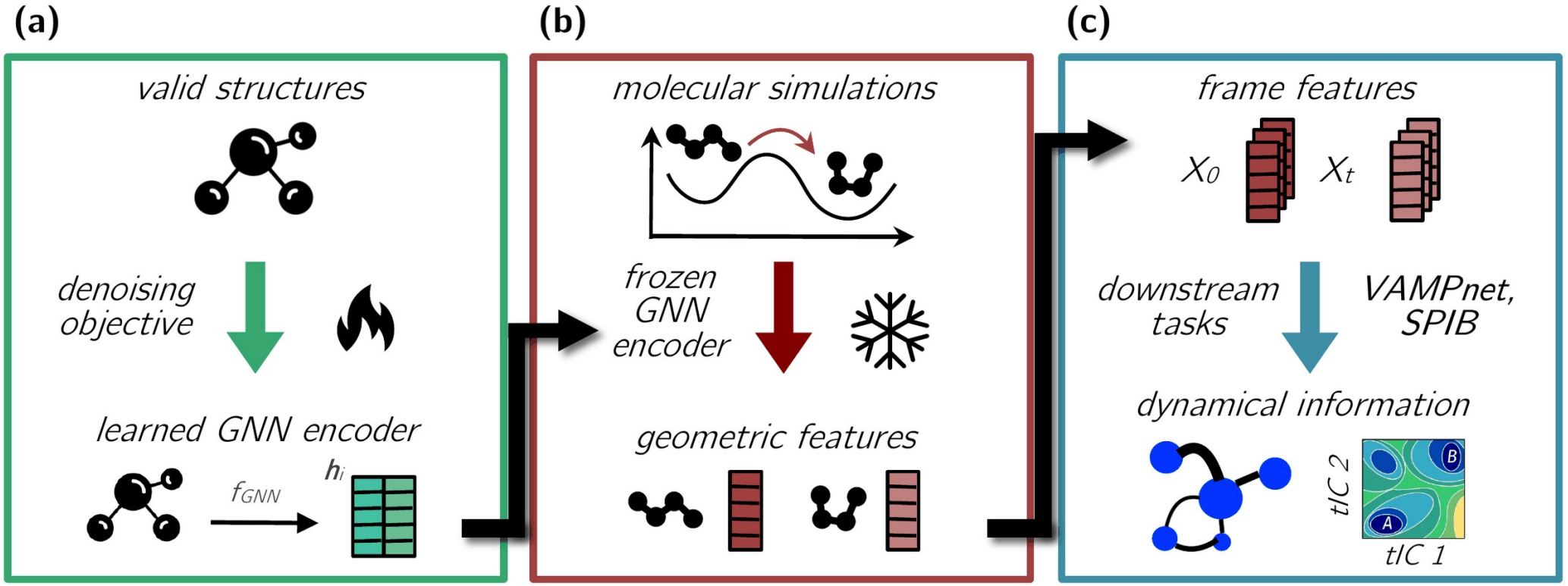
The states are distinguishable,
but transitions are missing.

Reinforcement learning: choose where to sample

A closed loop links representation, kinetics, and sampling



Pretrain once, then featurize every MD frame



Pretrain once

Denoising on 2.3 M conformations of 215 k molecules (OrbNet Denali)

Featurize frames

A frozen ViSNet encodes all heavy atoms; atoms pool into residue tokens

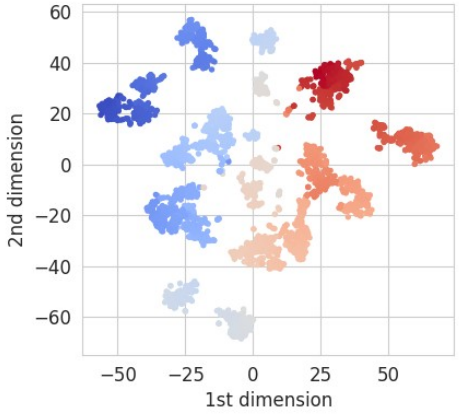
Train only the head

A token mixer (SubFormer, SubMixer) with a VAMP or SPIB objective

Freezing the geometric encoder makes all-atom kinetic learning fit on a single GPU.

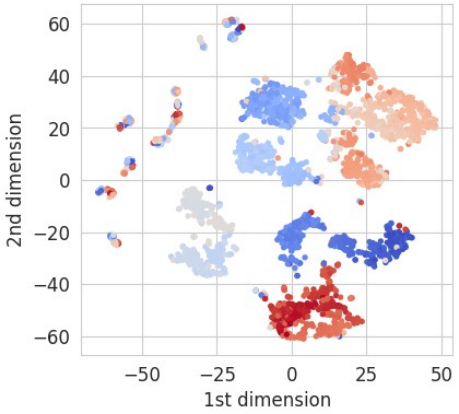
Pretrained features organize residues by local geometry

Angles (φ, ψ, χ)



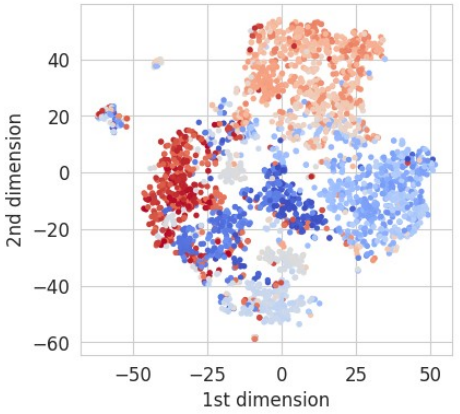
Clusters by construction: the reference.

Pretrained embedding



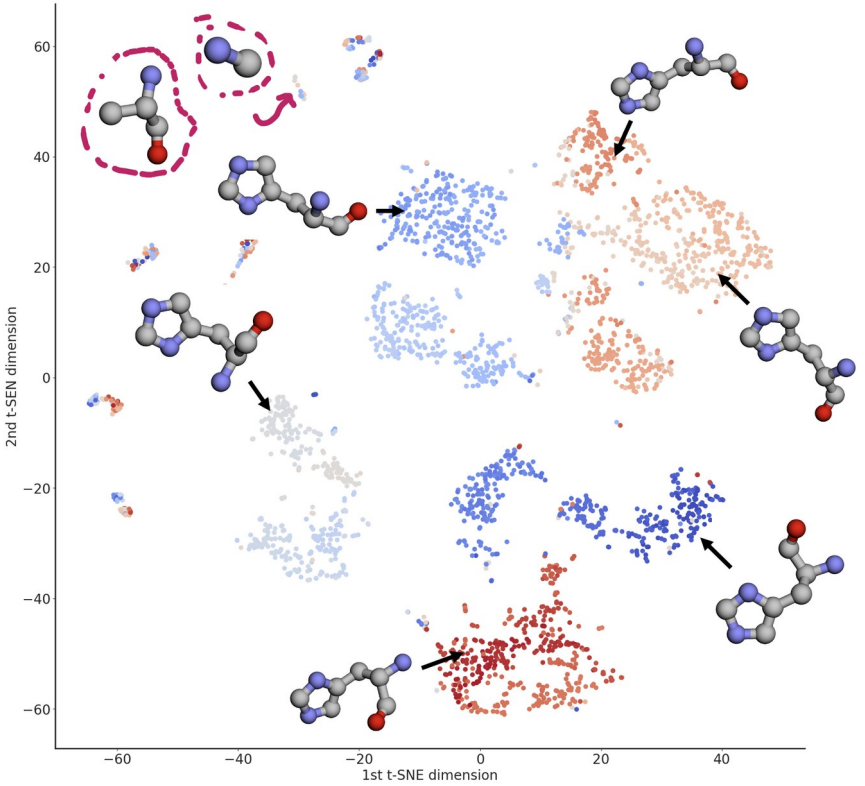
The same color groups stay together as compact clusters.

Untrained embedding



Groups overlap; the structure comes from pretraining.

The encoder never saw these angles: it was pretrained only to denoise coordinates of small molecules, yet residues with similar backbone and side-chain geometry map to nearby embeddings.

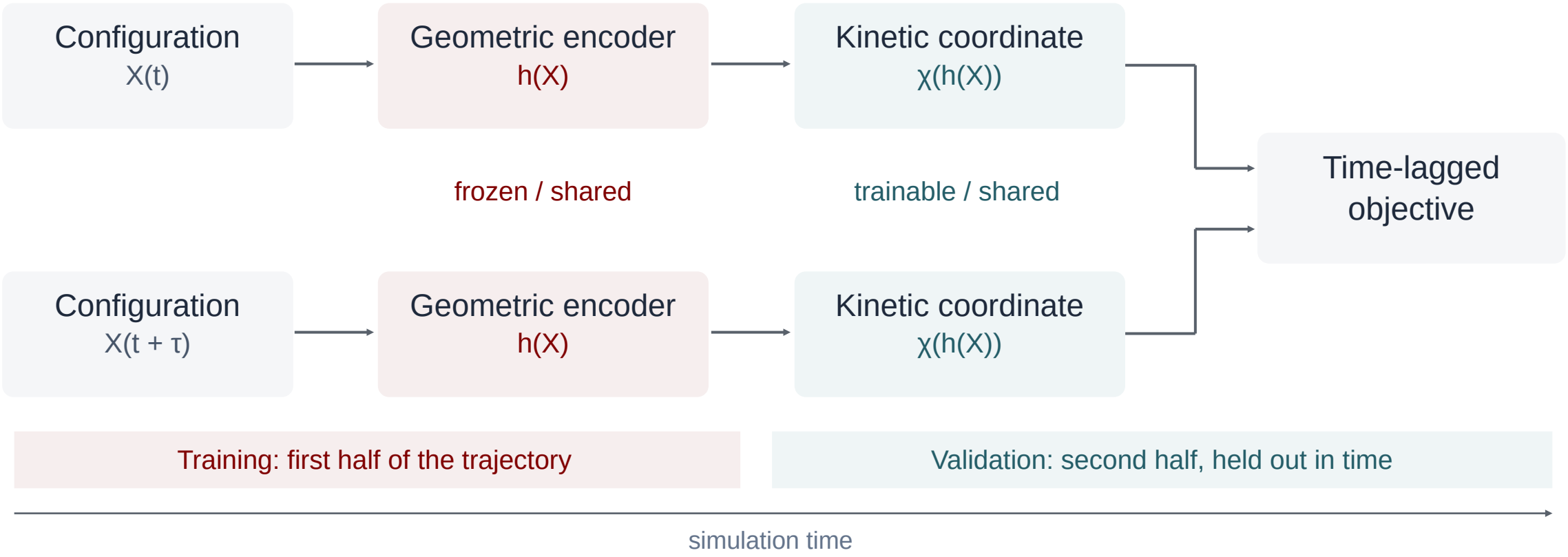


Each cluster is one side-chain conformation. Isolated points (circled) are broken fragments from dataset preprocessing errors.

Denoising pretraining alone yields features that resolve residue conformation, the input the kinetic models build on.

Pengmei et al., “Pushing the Limits of All-Atom Geometric Graph Neural Networks: Pre-Training, Scaling and Zero-Shot Transfer,” ICLR 2025, appendix. Histidine residues, HomologyTAPE validation set; color: 1D t-SNE of backbone and side-chain angles.

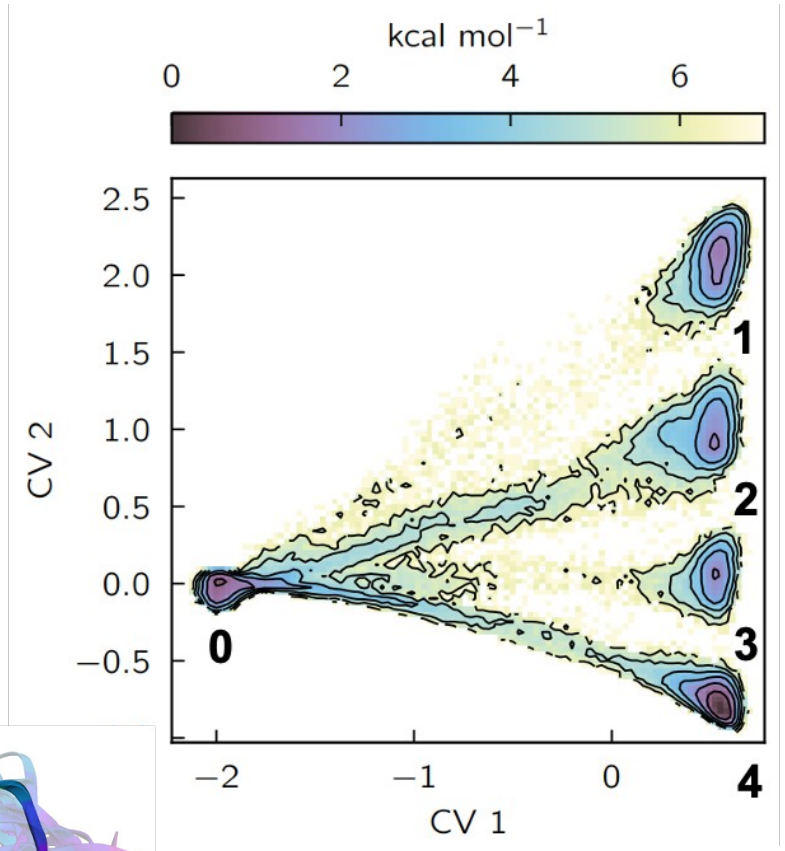
Time-lagged learning asks which distinctions persist

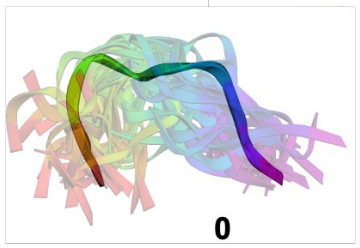


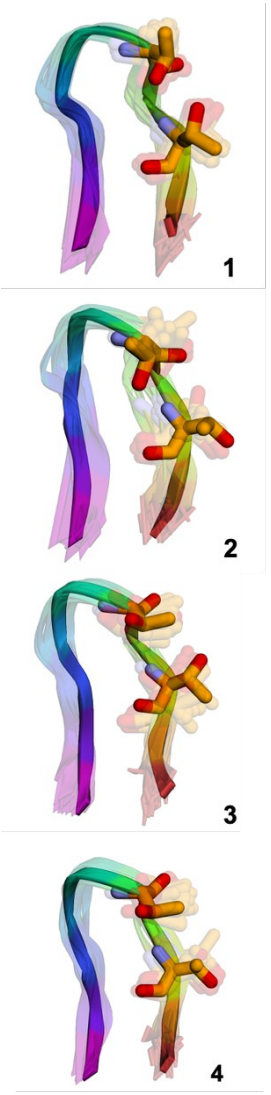
Split by time/trajectory, not by frame: random splits reward memorizing correlated neighbors.

Time-based split as in Pengmei et al., J. Chem. Phys. 162, 044107 (2025)

All-atom features resolve four folded substates







10-residue chignolin

106 μs unbiased trajectory at 340 K; all heavy atoms as input

Folding

CV 1 separates unfolded (0) from folded configurations.

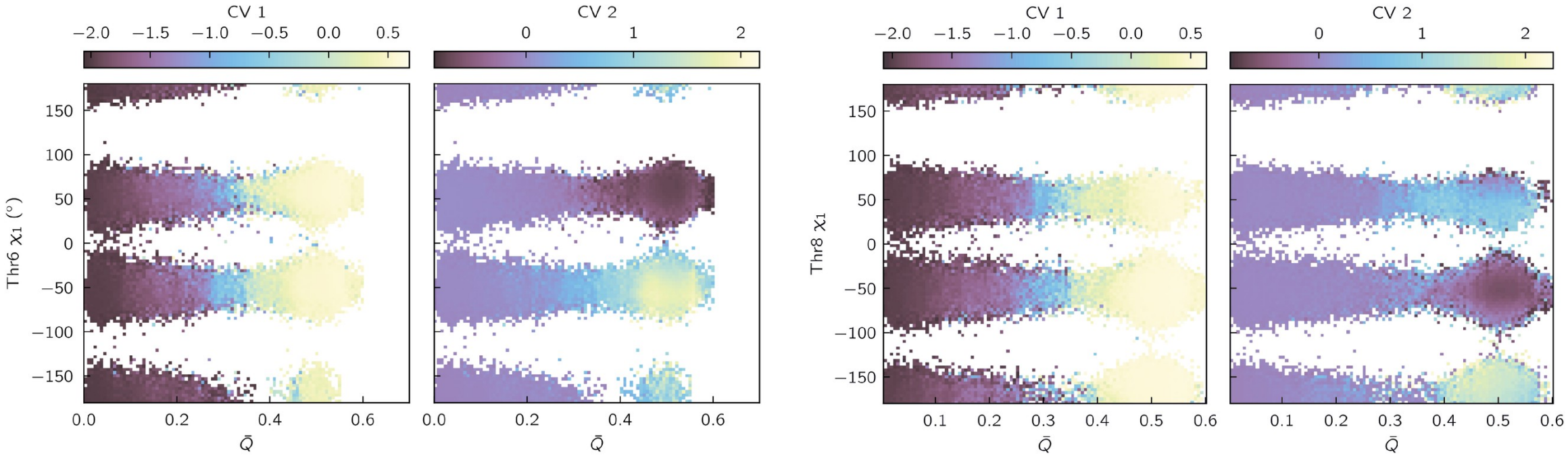
Folded substates

CV 2 splits the folded ensemble into four substates (1–4).

Thr6 and Thr8 rotamers define the four folded substates

Thr6

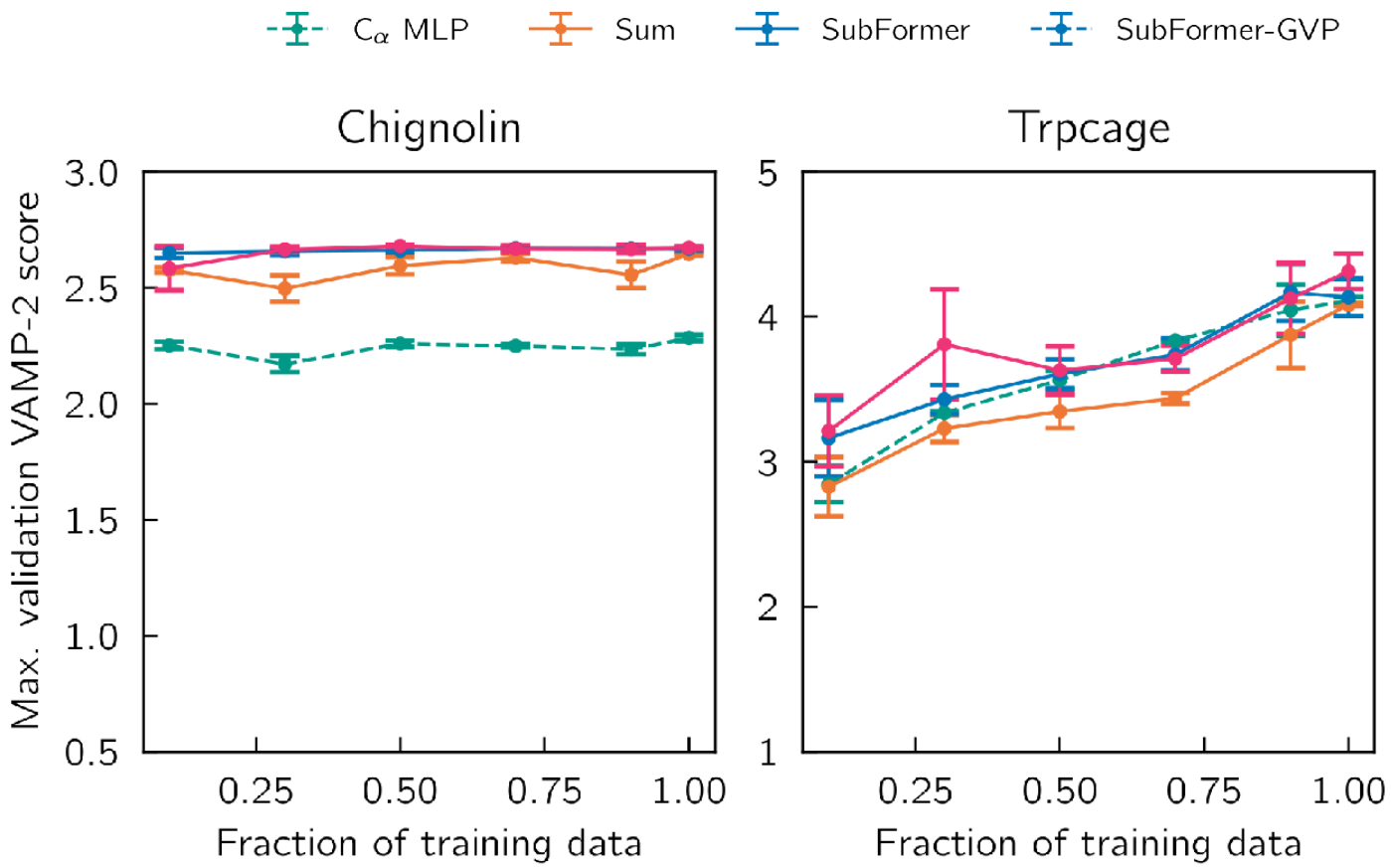
Thr8



x: fraction of native contacts Q̄ · y: χ_1 side-chain dihedral · color: mean CV value. CV 1 tracks Q̄; CV 2 tracks the rotamers.

No side-chain angle was hand-picked, and backbone-only inputs cannot see these substates.

All-atom features help most where side chains matter



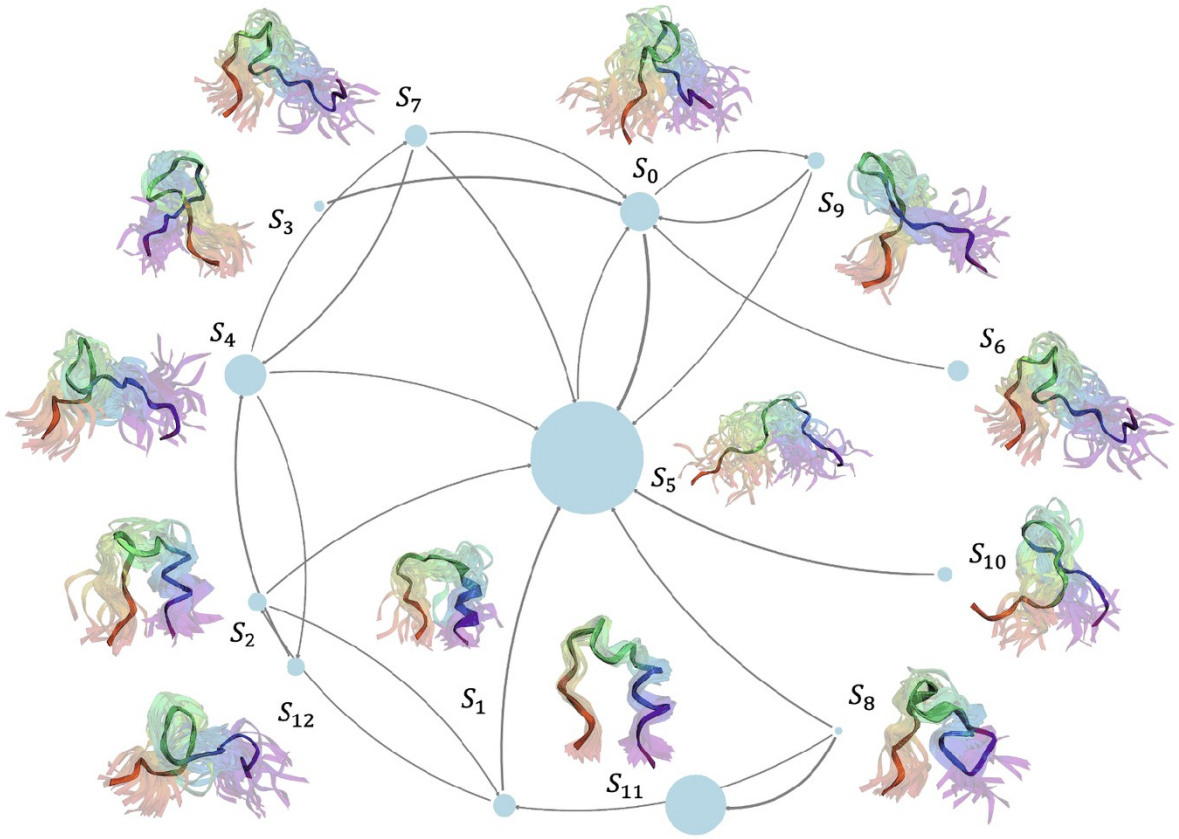
Chignolin: side chains matter

Every all-atom GNN variant beats the Cα-distance MLP.

Trp-cage: Cα distances suffice

Summed features score lowest; token mixers match the Cα MLP.

A learned state map can be read in molecular terms



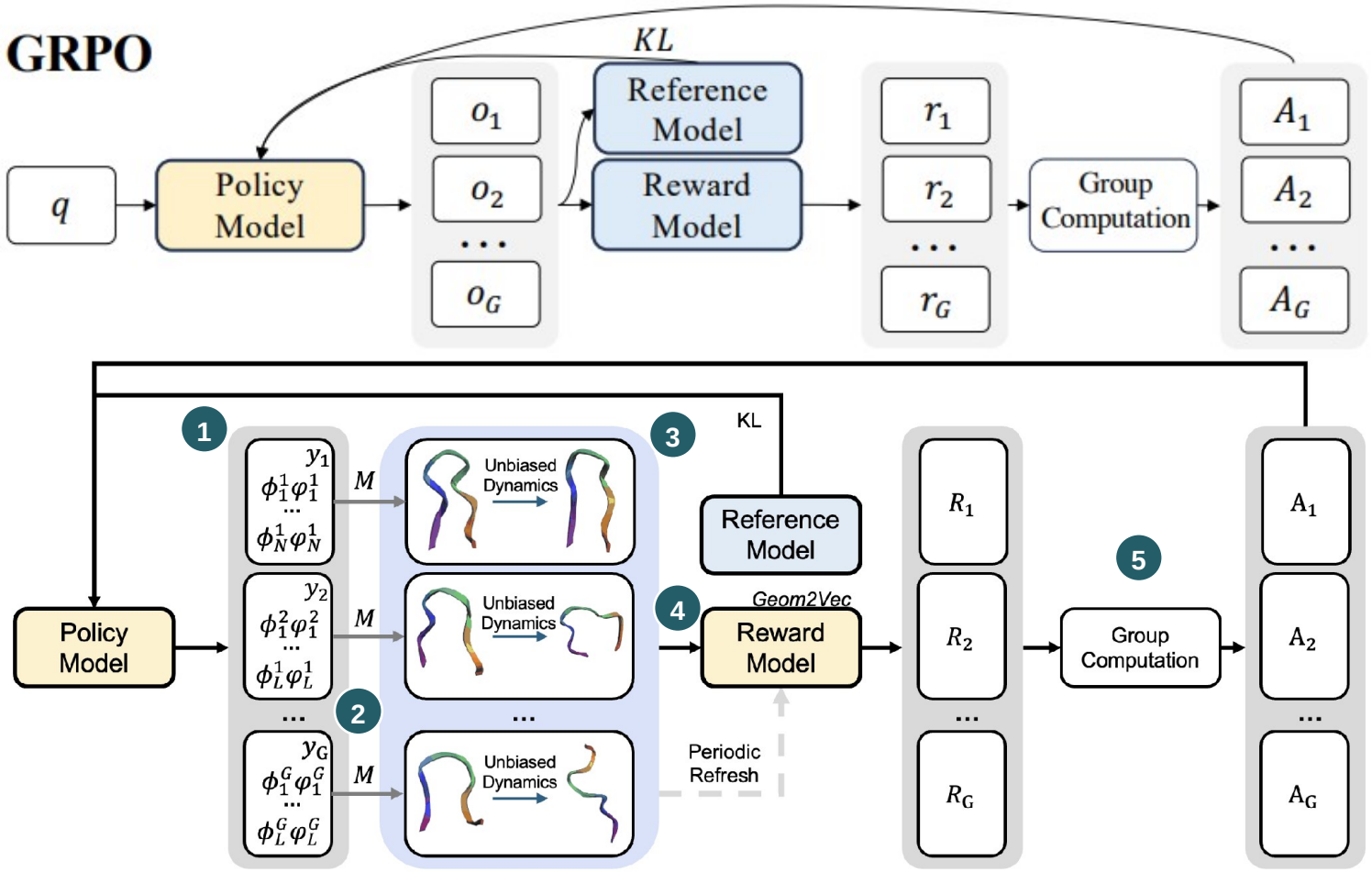
13 metastable states

S11 is folded; S1 and S2 keep the α-helix; S8 keeps the 3_{10}-helix.

An unfolded hub

S5 links every intermediate to the folded state; edges come from time-ordered transitions.

Adaptive sampling as policy optimization



Language models (GRPO)

A pretrained model generates a group of responses; a reward model scores each one; group-relative advantages update the model. GESPO keeps this update and derives the reward from simulated dynamics.

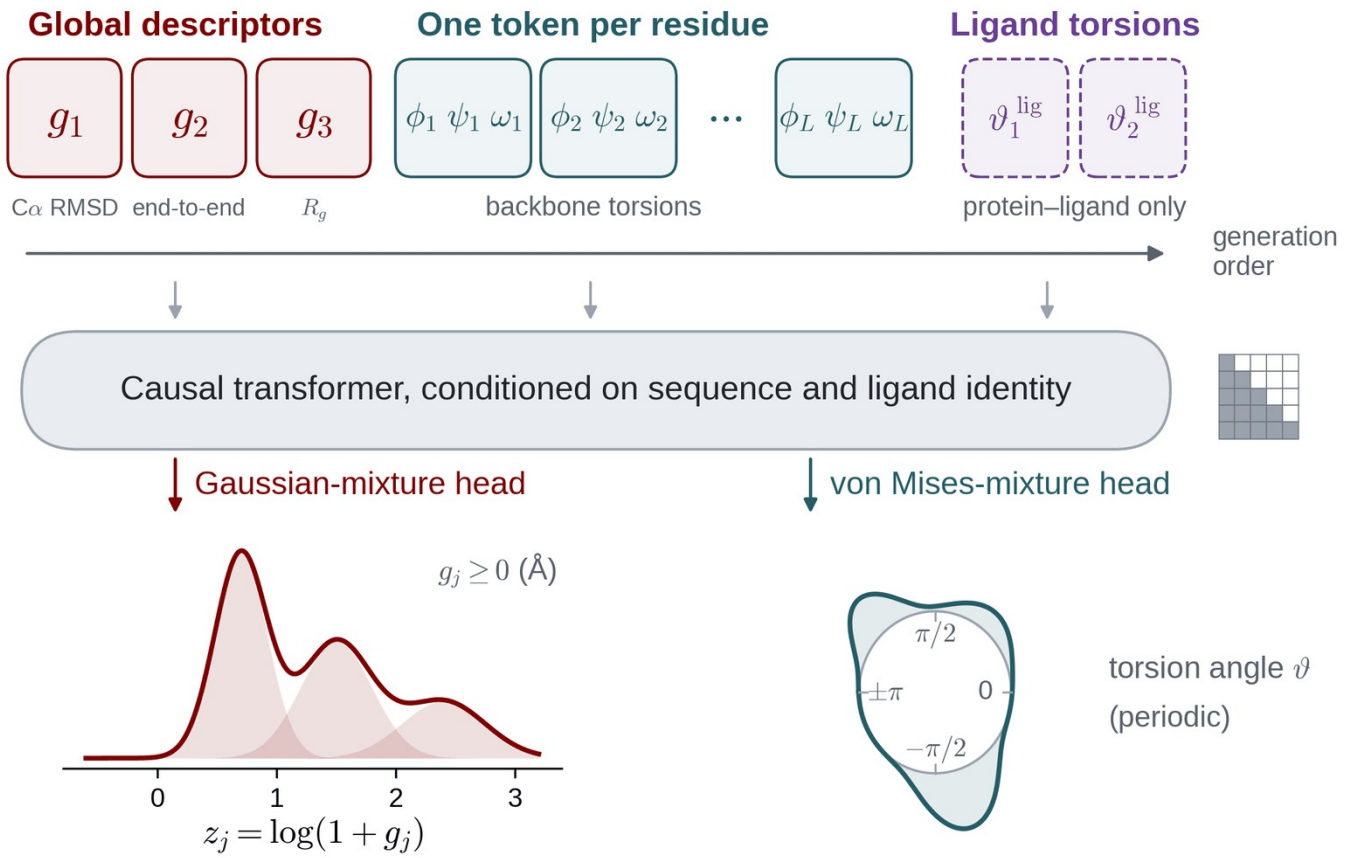
GESPO, one epoch

1. **Proposal:** the policy draws G initial conformations y_i.
2. **Realization ($\mathcal{M}$):** a nearby bank frame is restrained toward y_i, then released.
3. **Propagation:** short unbiased trajectories run in parallel.
4. **Reward:** the Geom2Vec kinetic model scores each trajectory.
5. **Update:** group-relative advantages drive a GRPO step.

The first RL-trained sampling policy for biomolecular MD; the dynamics remain unbiased.

Top: GRPO for language models (Shao et al., 2024). Bottom: GESPO loop, Pengmei et al., GESPO (manuscript, 2026)

The policy generates an initial conformation token by token



Autoregressive factorization

$$\pi_\theta(y) = \prod_{j=1}^{N_g} \pi_\theta(g_j \mid g_{<j}) \prod_{s=1}^{N_r} \prod_{c=1}^{3} \pi_\theta(\vartheta_{s,c} \mid g, \vartheta_{<(s,c)})^{m_{s,c}}$$

Global descriptors: Gaussian mixture

$$\pi_\theta(z_j \mid \cdot) = \sum_m w_m \mathcal{N}(z_j; \mu_m, \sigma_m^2), \qquad z_j = \log(1 + g_j)$$

Torsions: von Mises mixture

$$\pi_\theta(\vartheta \mid \cdot) = \sum_m w_m \, \mathrm{VM}(\vartheta; \mu_m, \kappa_m), \qquad \mathrm{VM} = \frac{e^{\kappa \cos(\vartheta - \mu)}}{2\pi I_0(\kappa)}$$

Exact log-likelihood

$$\log \pi_\theta(y) = \sum_j \log \pi_\theta(g_j \mid \cdot) + \sum_{s,c} m_{s,c} \log \pi_\theta(\vartheta_{s,c} \mid \cdot)$$

Every conditional is a normalized density, so log $\pi_\theta(y)$ is exact.

Geom2Vec is the kinetic model inside GESPO

Geom2Vec

Frozen all-atom encoder

↓ residue (and ligand) tokens

↓ token merging (trypsin)

↓ VAMP projector χ, refit each epoch

Not Geom2Vec: the policy proposes torsions and global descriptors; realization matches frames by torsions; alanine dipeptide uses (φ, ψ) directly.

Reward: scores each trajectory

Leave-one-out VAMP-2 contribution
Change in S when trajectory i is removed

Coverage bonus
Visits to sparsely sampled bins in the first two slow coordinates

Transition information gain
New transitions among k-means states in the slow coordinates

Analysis: interprets the data

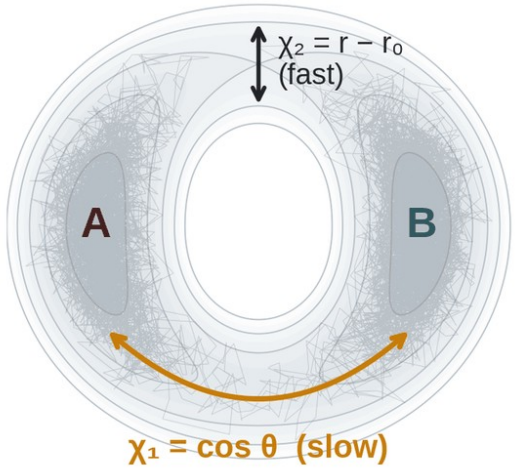
SPIB metastable states
Trp-cage: 23 states. Trypsin: 7 receptor macrostates

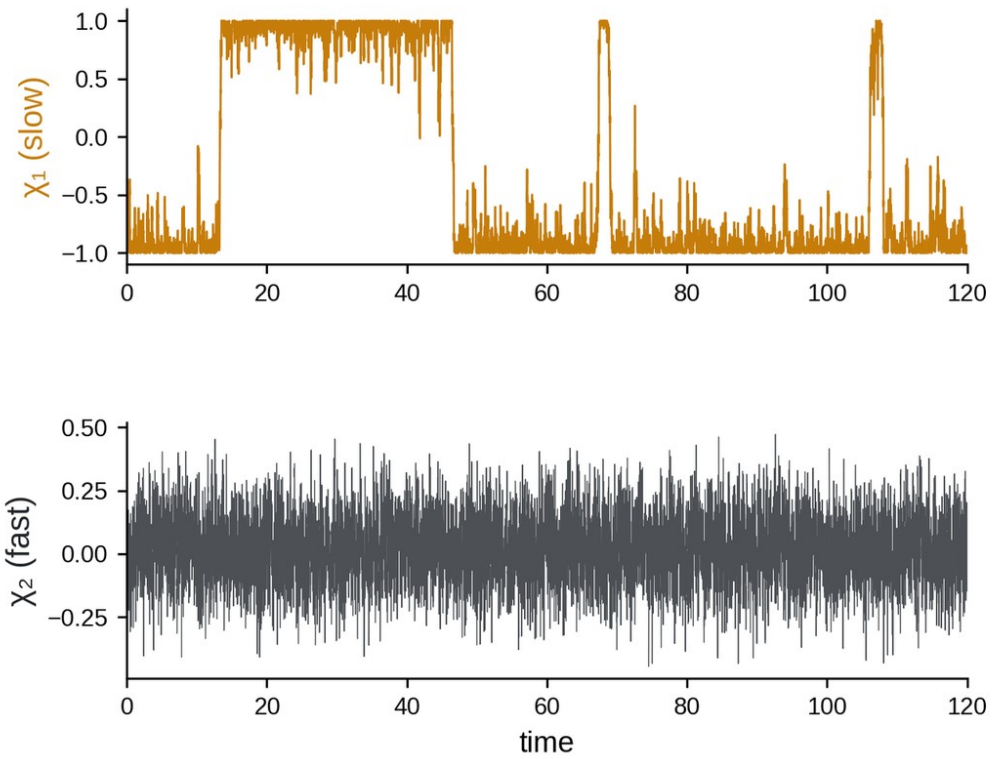
DGA committors
Basis functions built on the SPIB states

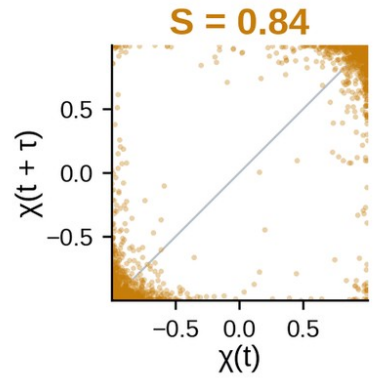
Reactive flux and exchange
Networks over the SPIB macrostates

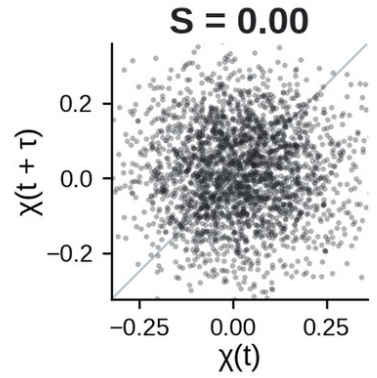
What counts as kinetic information, for the reward and for the analysis, is defined in Geom2Vec coordinates.

VAMP-2 rewards features that remain correlated after a lag









$$S(\mathcal{D}) = \left\| C_{00}^{-1/2}\, C_{0\tau}\, C_{\tau\tau}^{-1/2} \right\|_F^2 = \sum_k \sigma_k^2$$

C_{00}, $C_{\tau\tau}$: instantaneous covariances; $C_{0\tau}$: lagged covariance, from pairs within each trajectory. For one feature, S is the squared lag correlation.

The reward asks how much each trajectory contributes to this score.

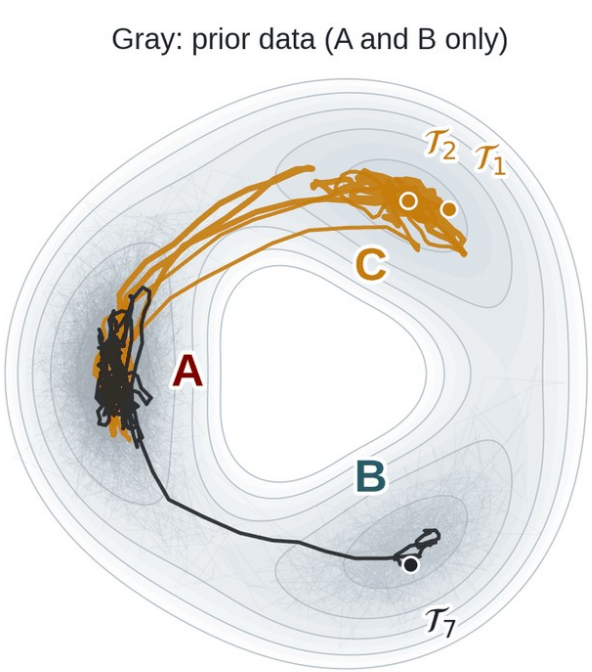
Toy model from slide 3 (landscape 1), 24 trajectories, τ = 1. In GESPO, χ is learned from Geom2Vec features and scored the same way.

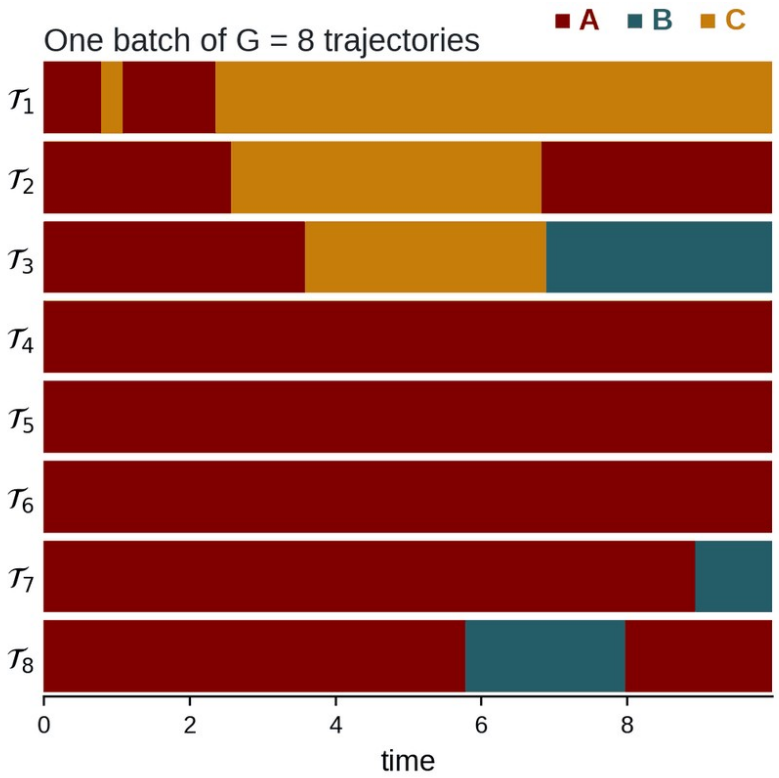
Leave-one-out credit, compared within a batch

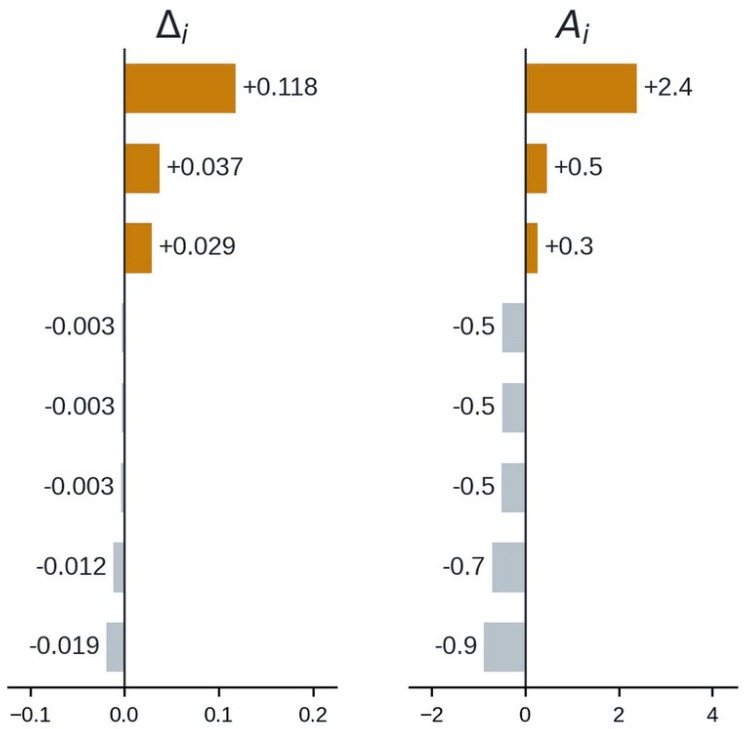
$$\Delta_i = S(\mathcal{D}) - S(\mathcal{D} \setminus \{\mathcal{T}_i\})$$

$$A_i = (R_i - \langle R \rangle) / s_R$$

χ held fixed; means and covariances recomputed. Advantages standardize within the batch.







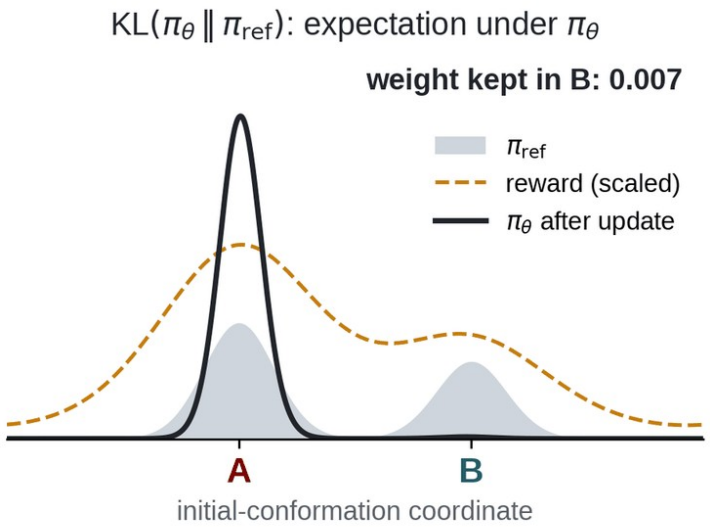
The trajectory that settles in the new state C earns the largest advantage.

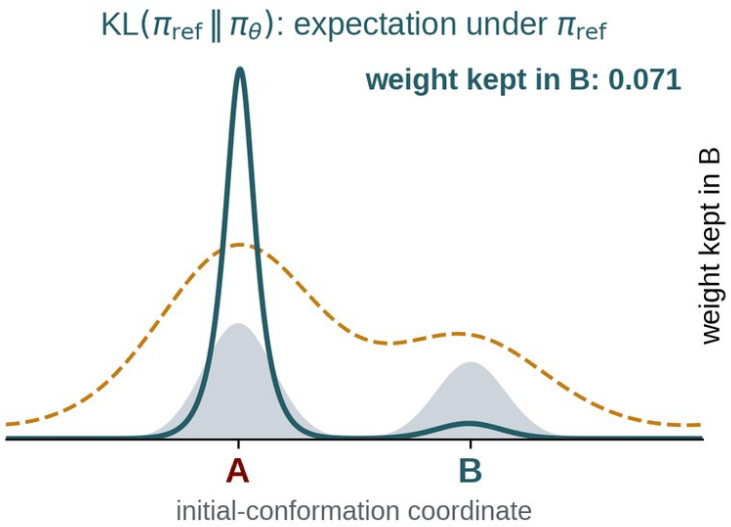
Three-state toy model; χ = (cos θ, sin θ) held fixed; $\mathcal{D}$ = 40 prior trajectories (A, B only) plus the batch; T = 10, τ = 1. Here $R_i = \Delta_i$; GESPO adds coverage and transition terms.

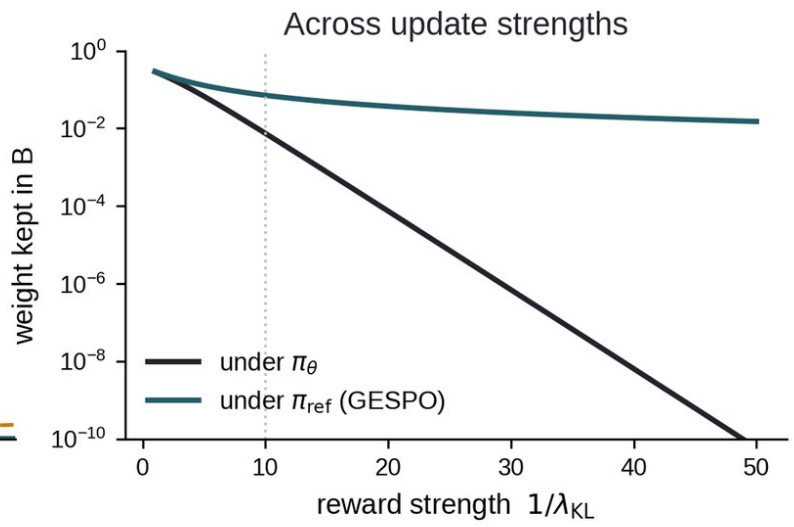
A coverage-preserving KL term regularizes the update

$$\max_\theta \; \mathbb{E}_{y\sim\pi\theta}[R(y)] \; - \; \lambda_{KL}\, D_{KL}(\pi_{ref} \,\|\, \pi_\theta) \; + \; \lambda_H\, H(\pi_\theta)$$

Optimized with the GRPO clipped surrogate on $\rho_i = \pi_\theta(y_i) / \pi_{old}(y_i)$ and group advantages A_i; $\pi_{ref} = \pi_{old}$, resynchronized each epoch.







Under π_θ, leaving B costs almost nothing: its weight decays as $e^{-\Delta R/\lambda}$.

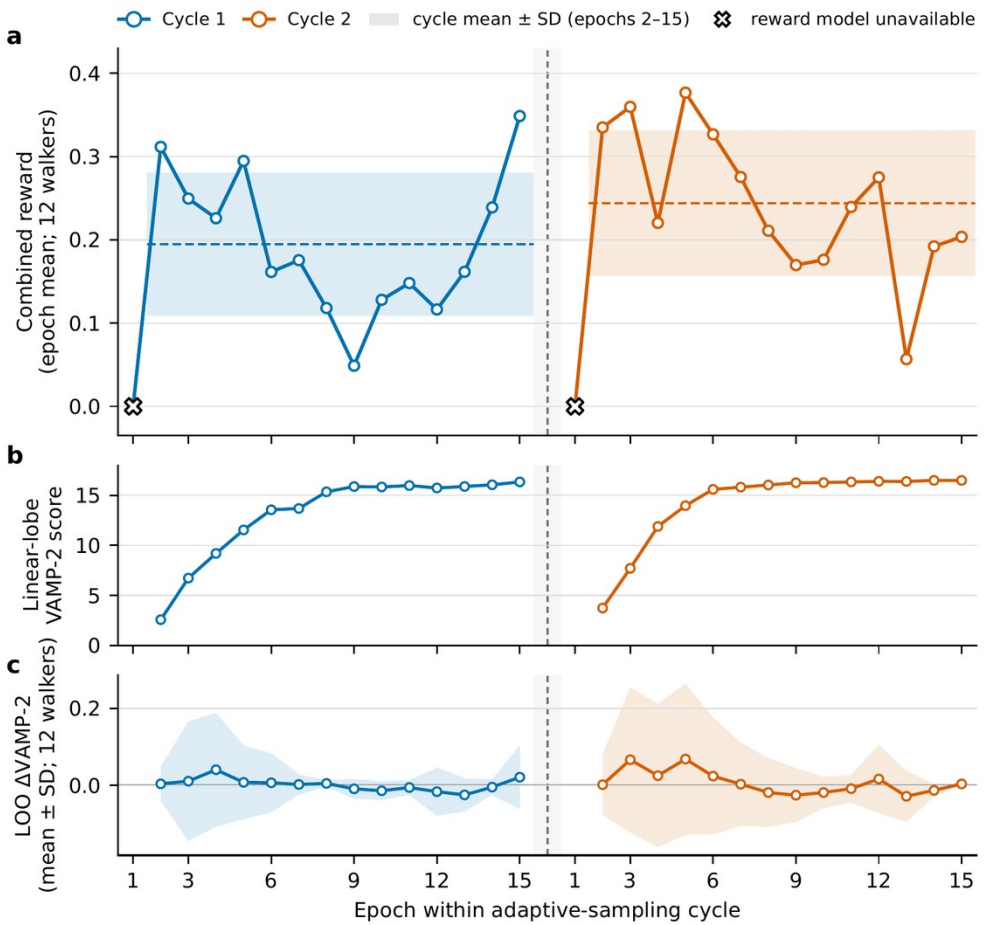
Under π_{ref}, leaving B is penalized: its weight decays only in proportion to λ.

The ordering matters: a sampling distribution that contracts each epoch eventually concentrates on a single basin.

Taken under π_{ref}, the KL term keeps weight on every basin the reference samples.

Toy example: exact optimum of the regularized objective for a fixed reference (λ_{KL} = 0.1 in the left panels). GRPO: Shao et al. (2024); GESPO: Pengmei et al. (manuscript, 2026).

Diagnostics: the kinetic model converges within each round



(a) Combined reward

Epoch mean over 12 trajectories: the leave-one-out contribution plus coverage and transition terms.

(b) VAMP-2 score of the refitted model

Refitted before every epoch; rises and plateaus within each round.

(c) Leave-one-out contribution Δ_i

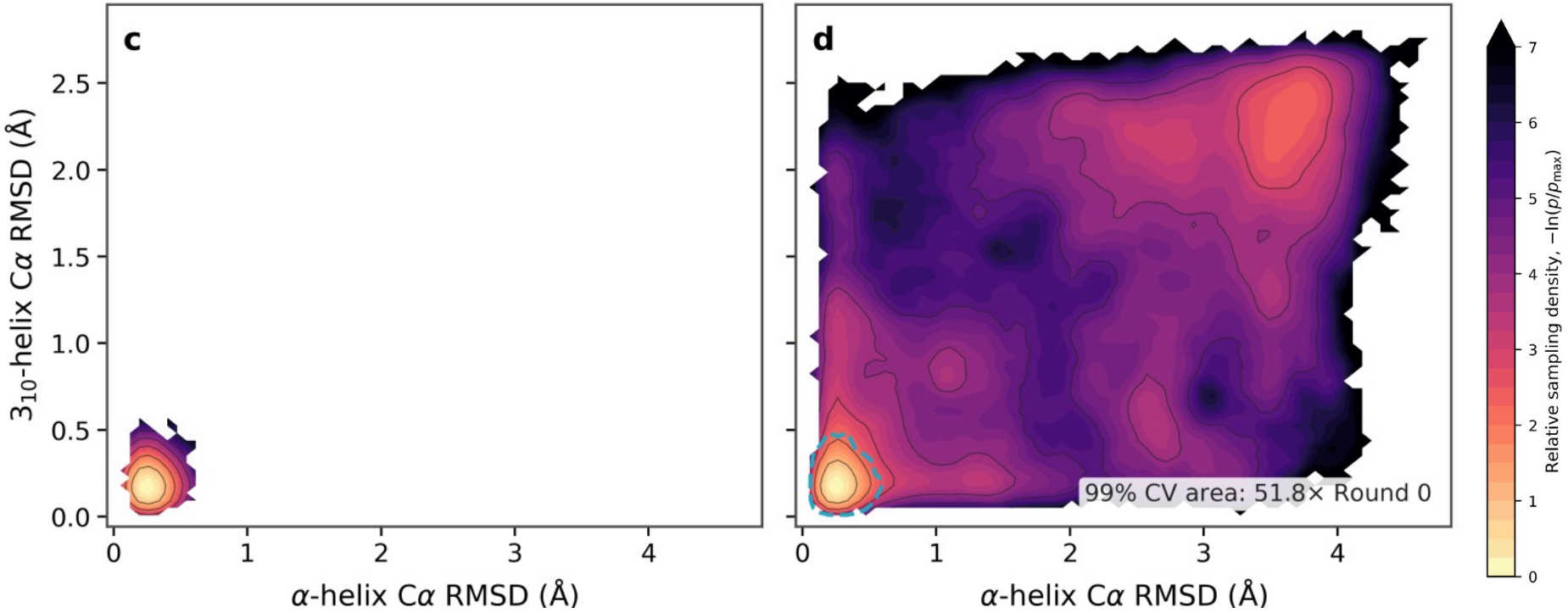
Mean near zero with a wide spread across trajectories, as in the toy example: credit is relative within each batch.

The reward ranks trajectories within a batch; the kinetic model improves across epochs.

Trp-cage: sampled area grows 52-fold beyond the initial data

Initial data (round 0): 4.8 ns

All rounds: 115.2 ns



51.8×

area of the sampled region vs. the initial data (99% contour)

115.2 ns

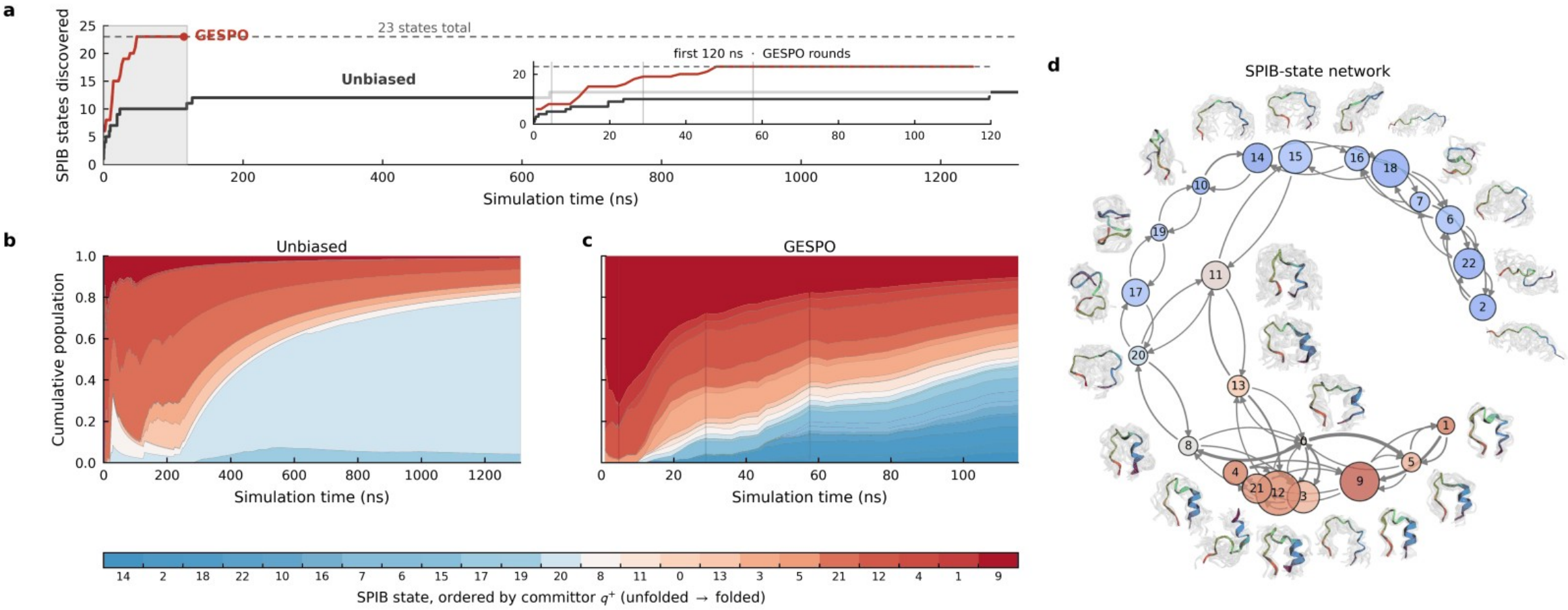
cumulative production

12 × 200 ps

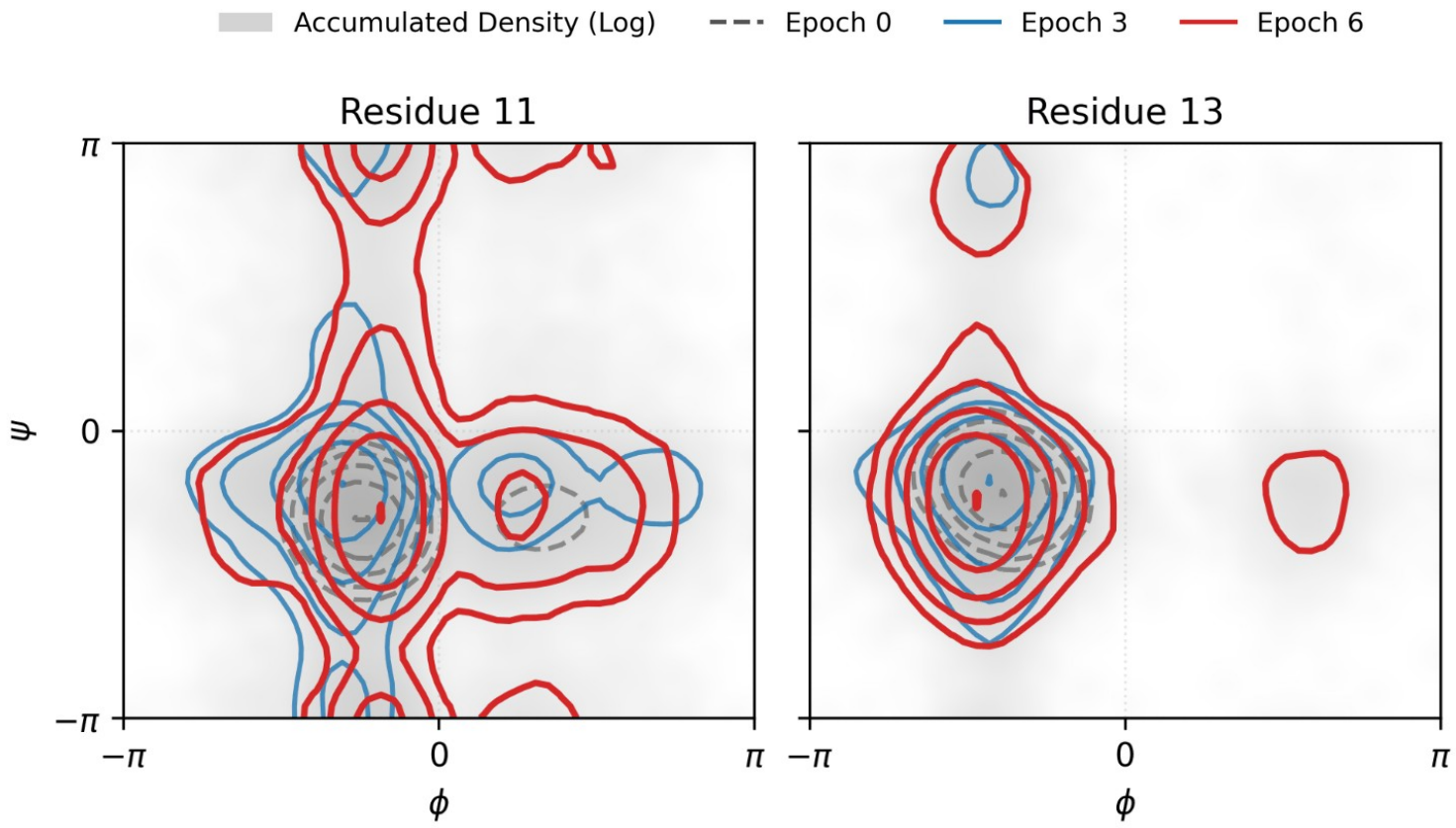
trajectories per epoch

Pengmei et al., GESPO (manuscript, 2026). Color: relative sampling density, $-\ln(p/p_{max})$, not a free energy. Dashed contour: extent of the round-0 data.

All 23 states in ~50 ns; unbiased MD reaches 12 in 1.3 µs



The sampling distribution broadens rather than collapsing



Epochs 0 → 3 → 6

The torsion marginals widen into φ > 0 and toward ψ = ±π, regions the initial policy did not populate.

Consistent with the KL ordering

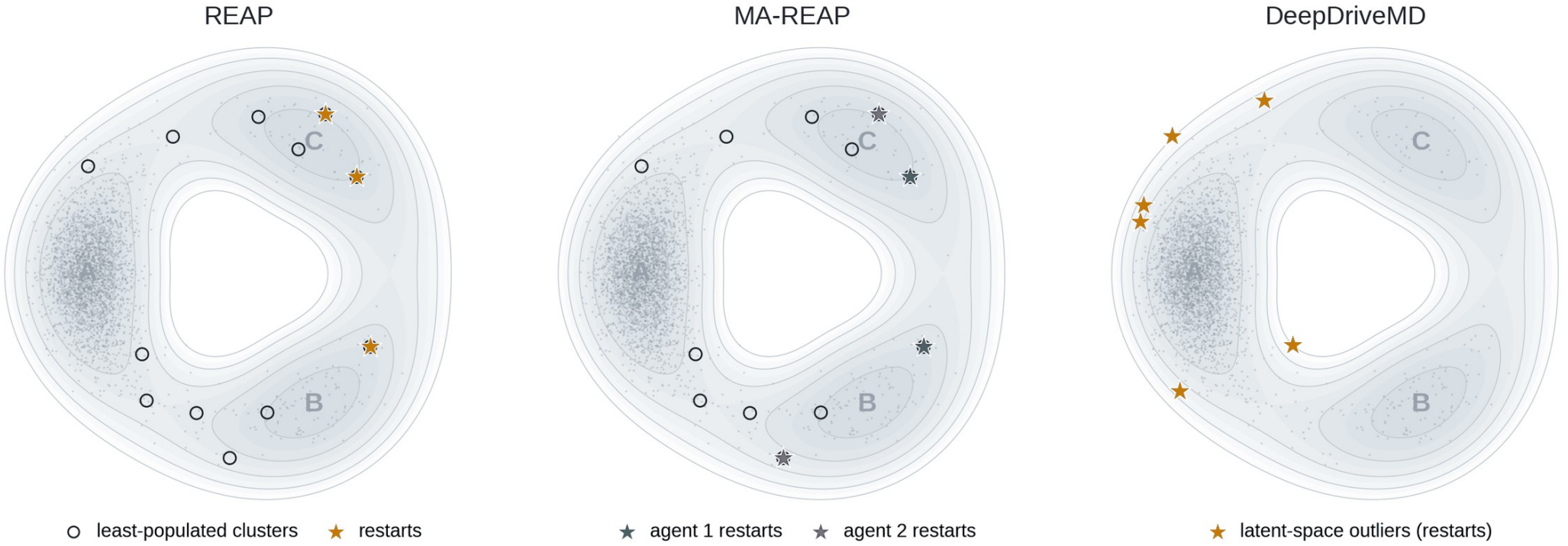
No contraction onto a subset of the initial distribution (slide 15).

Marginals do not show the joint distribution over all residues.

The update adds new torsion regions.

Pengmei et al., GESPO (manuscript, 2026). Policy torsion marginals for Trp-cage residues 11 and 13; all ten residues in backup.

Established adaptive samplers rank candidates by novelty



REAP Shamsi et al., 2018

Cluster the data; among the least-populated clusters, restart from those farthest along weighted CVs. The weights are refit each round.

MA-REAP Kleiman & Shukla, 2022

Several REAP agents, each with its own CV weights, share the sampled data and choose restarts jointly.

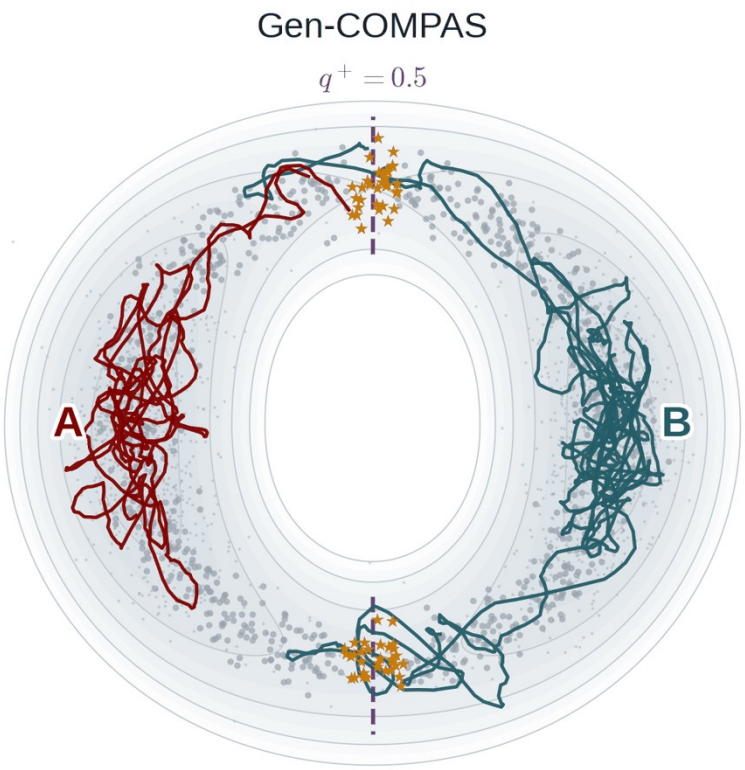
DeepDriveMD Lee et al., 2019

An autoencoder embeds the structures; restarts come from outliers in its latent space.

Each rule scores candidates from structures alone, before any trajectory is run.

Toy illustration on the three-state ring: 40 short trajectories from A; clustering and REAP reward computed on these data; the DeepDriveMD latent space is drawn in the toy coordinates.

Gen-COMPAS shoots trajectories from the separatrix



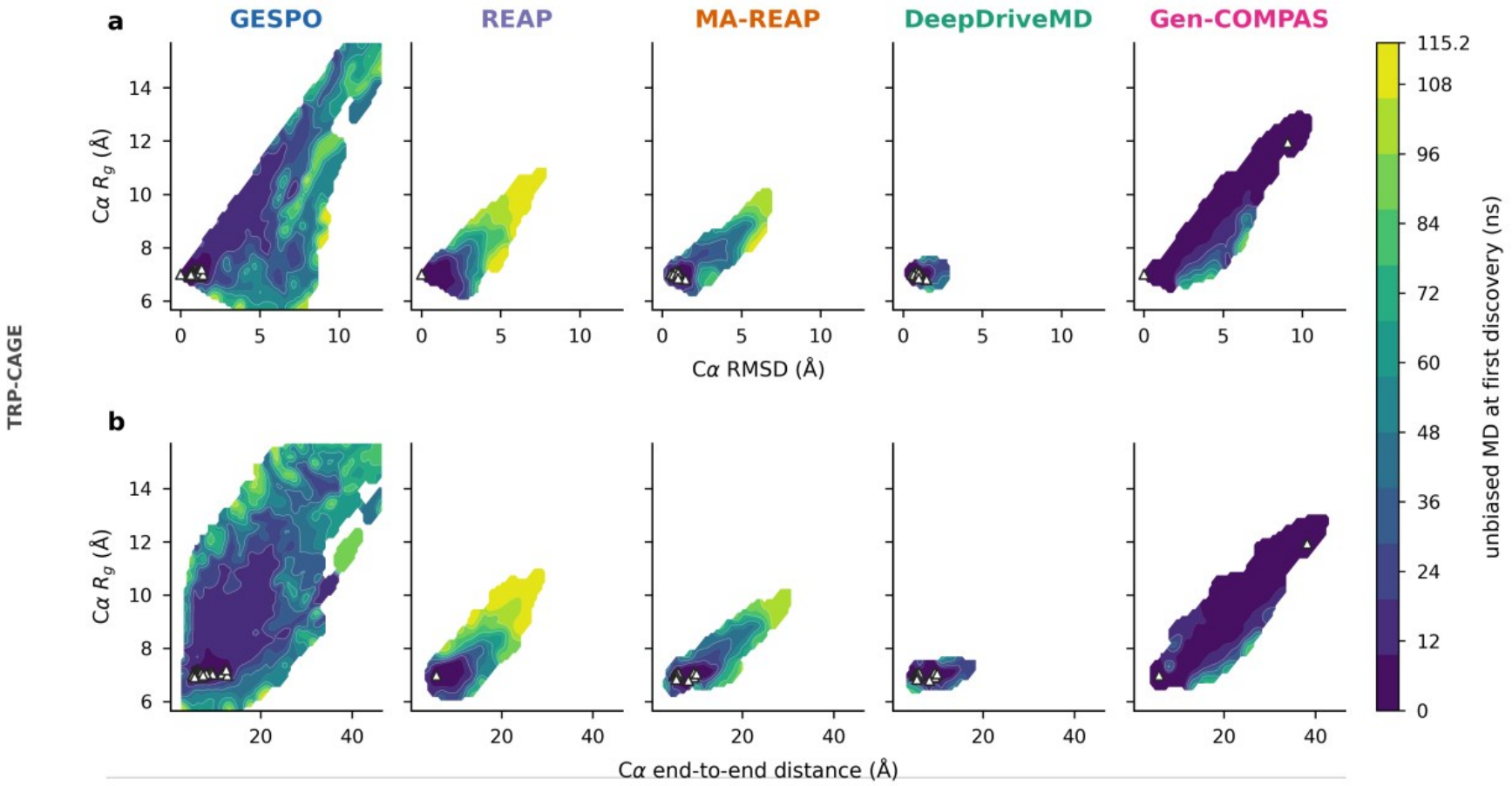
1. Train a diffusion model on short unbiased trajectories from reactant A and product B.
2. Generate intermediates; keep those with a learned committor $q^+ \approx 0.5$ (stars).
3. Shoot unbiased trajectories from them; refit both models each round.

Method	Chooses where to sample by	Requires
REAP, MA-REAP	Novelty along weighted CVs	Collective variables
DeepDriveMD	Outliers in a learned latent space	An autoencoder
Gen-COMPAS	Generated structures near $q^+ = 0.5$	Reactant and product states
GESPO	**A policy trained on trajectory-level rewards**	**No predefined end states**

GESPO trains a generative policy on trajectory outcomes; the others rank existing structures.

Gen-COMPAS: Tang et al., arXiv:2510.24979 (2025). Toy illustration on the slide 3 landscape; generated structures and committor are schematic (1D angular committor).

At equal budget, GESPO covers the widest region



GESPO

Reaches Cα RMSD 12 Å and end-to-end distances near 50 Å; still discovering late in the budget.

REAP, MA-REAP

Stop near 8–9 Å RMSD and 30 Å end-to-end distance.

DeepDriveMD

Stays within 3 Å of the folded state.

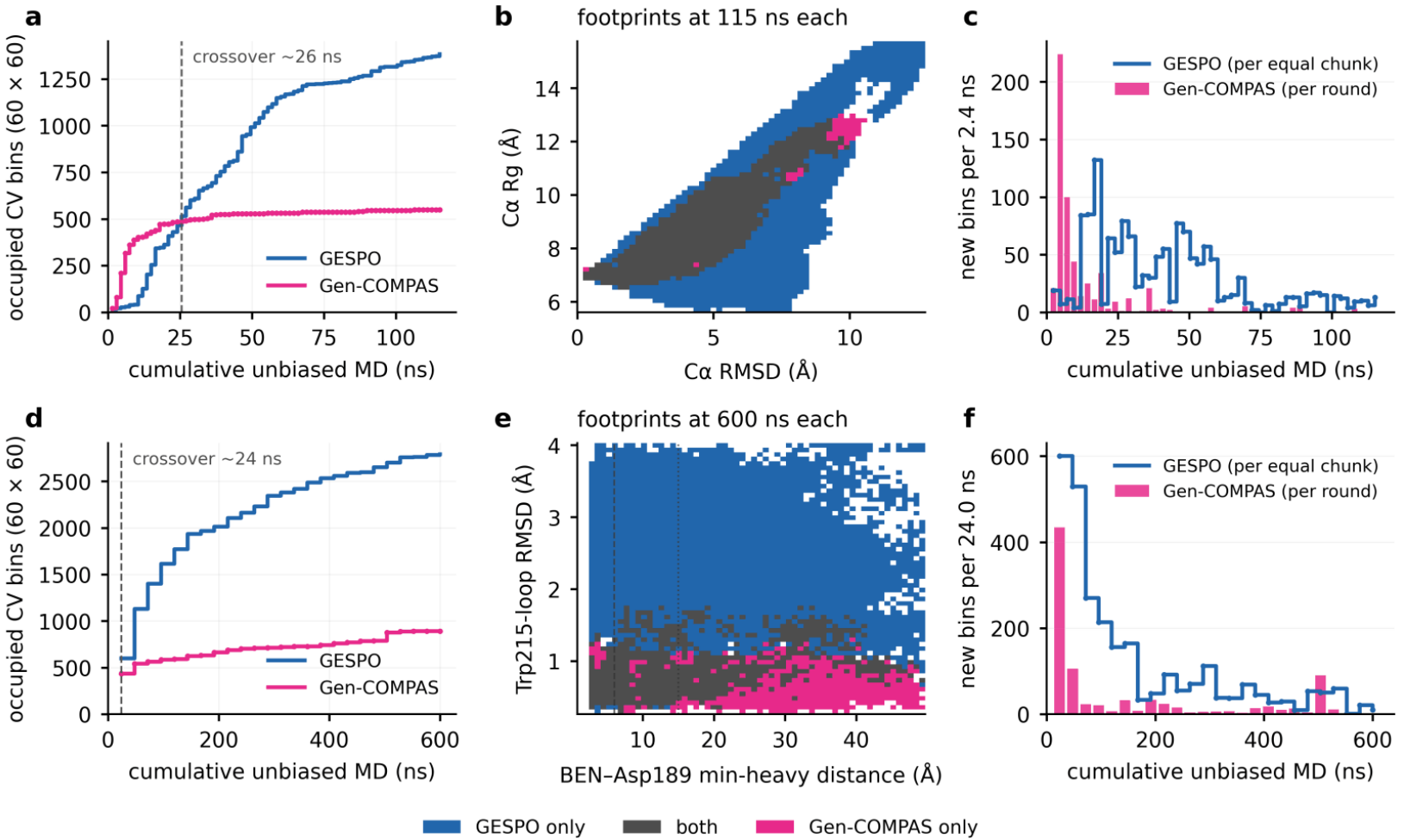
Gen-COMPAS

Extends to ~10 Å, but along one narrow band, reached early.

Same system and budget: only the choice of where to sample differs.

Pengmei et al., GESPO (manuscript, 2026). 115.2 ns per method; same engine, force field and prepared system. Color: budget at first discovery; triangles: initial conformations.

Gen-COMPAS plateaus early; GESPO keeps discovering



Trp-cage, 115 ns

≈2.5× more occupied bins (≈1,380 vs. ≈550).

Trypsin, 600 ns

≈3× more occupied bins (≈2,800 vs. ≈900).

Early lead, then plateau

Gen-COMPAS leads for the first ~25 ns; most of its new bins come in its first rounds (c, f).

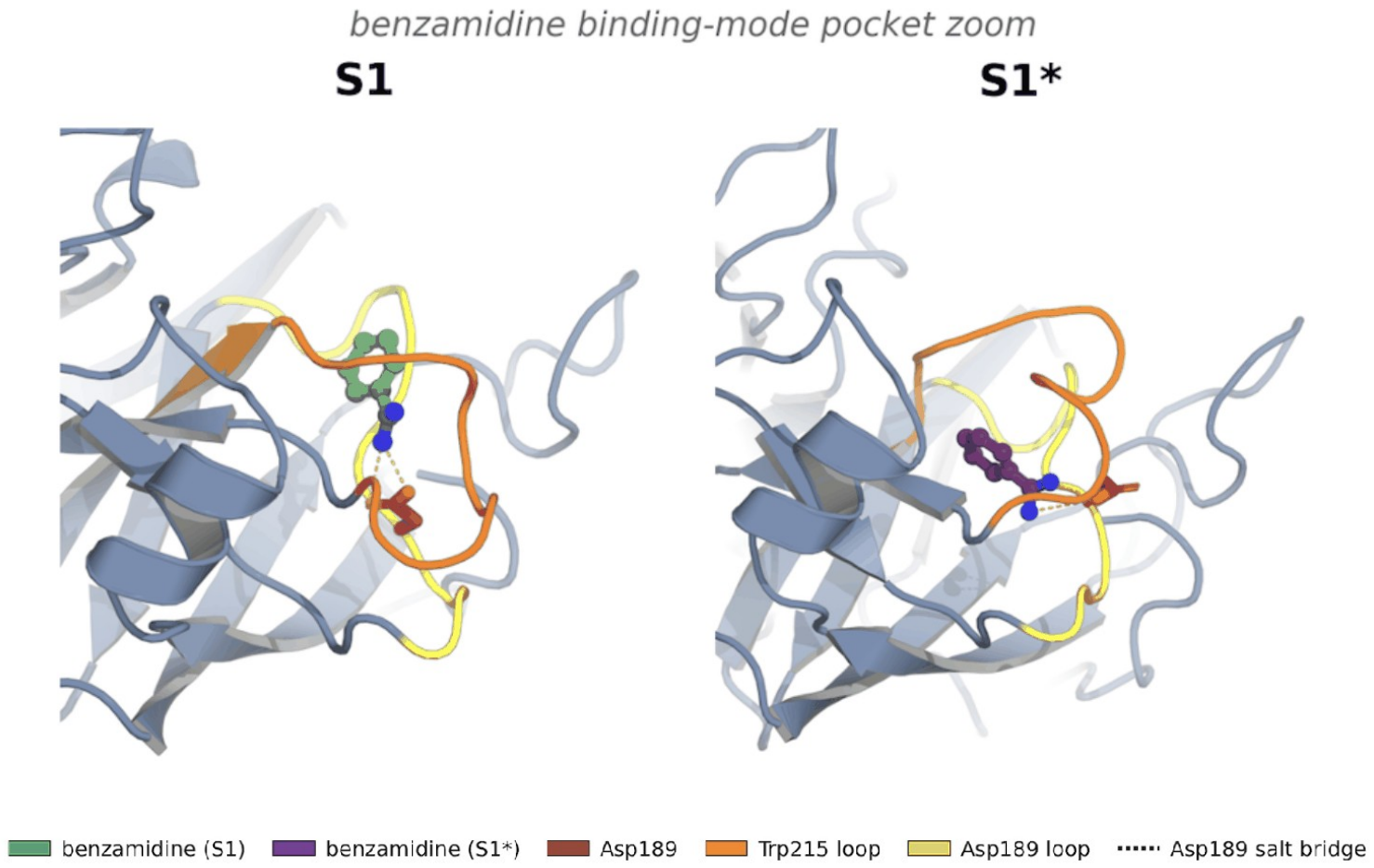
Footprints (b, e)

Gen-COMPAS covers a band between its end states; GESPO extends across both coordinates.

Sampling between known end states saturates; a learned policy continues to find new regions.

Pengmei et al., GESPO (manuscript, 2026). Occupied bins on a 60 × 60 CV grid. Top: Trp-cage, 115 ns per method; bottom: trypsin–benzamidine, 600 ns per method.

Trypsin–benzamidine: resolving ligand and receptor together



The question

Through which receptor conformations does benzamidine bind and leave?

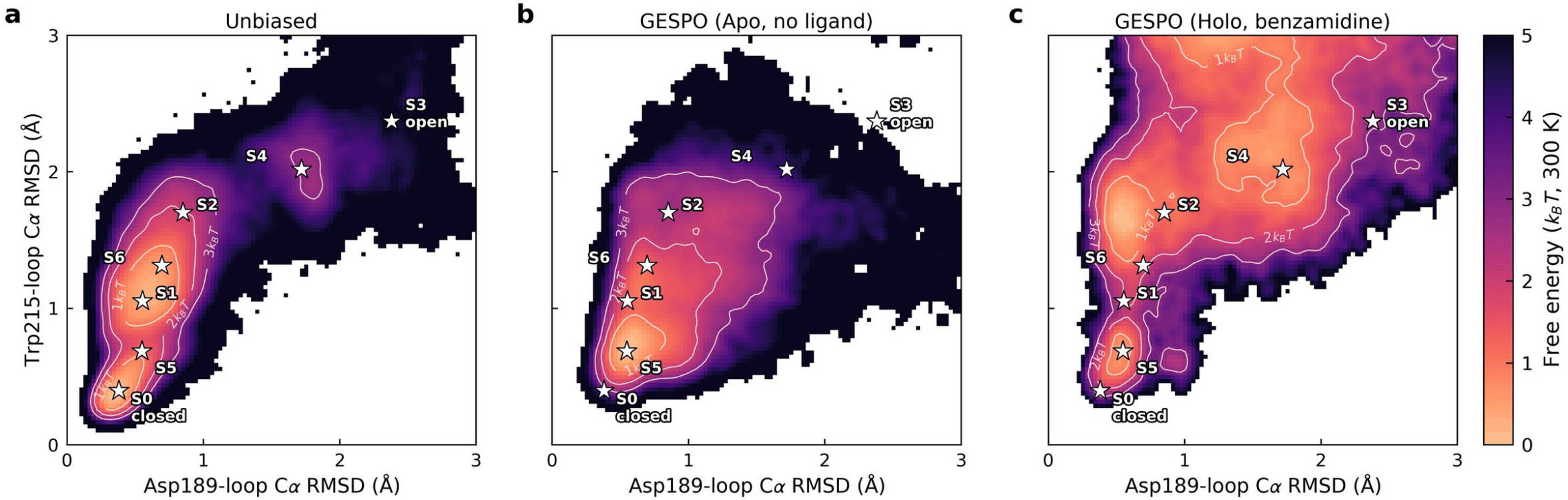
Production

12 trajectories × 2 ns × 30 epochs = 720 ns of unbiased dynamics

Initial data: 100 ns of metadynamics, used only to pretrain the policy and fill the conformation bank; excluded from all kinetic estimates.

Structures as on slide 2; Pengmei et al., GESPO (manuscript, 2026)

With and without ligand, GESPO sampling follows the physics



(a) Unbiased reference (holo), 100 µs

Seven receptor states (stars); the open states lie several k_BT above the main band.

(b) GESPO, apo

Same protocol without benzamidine: covers the reference receptor states, and the Trp215 gate still opens.

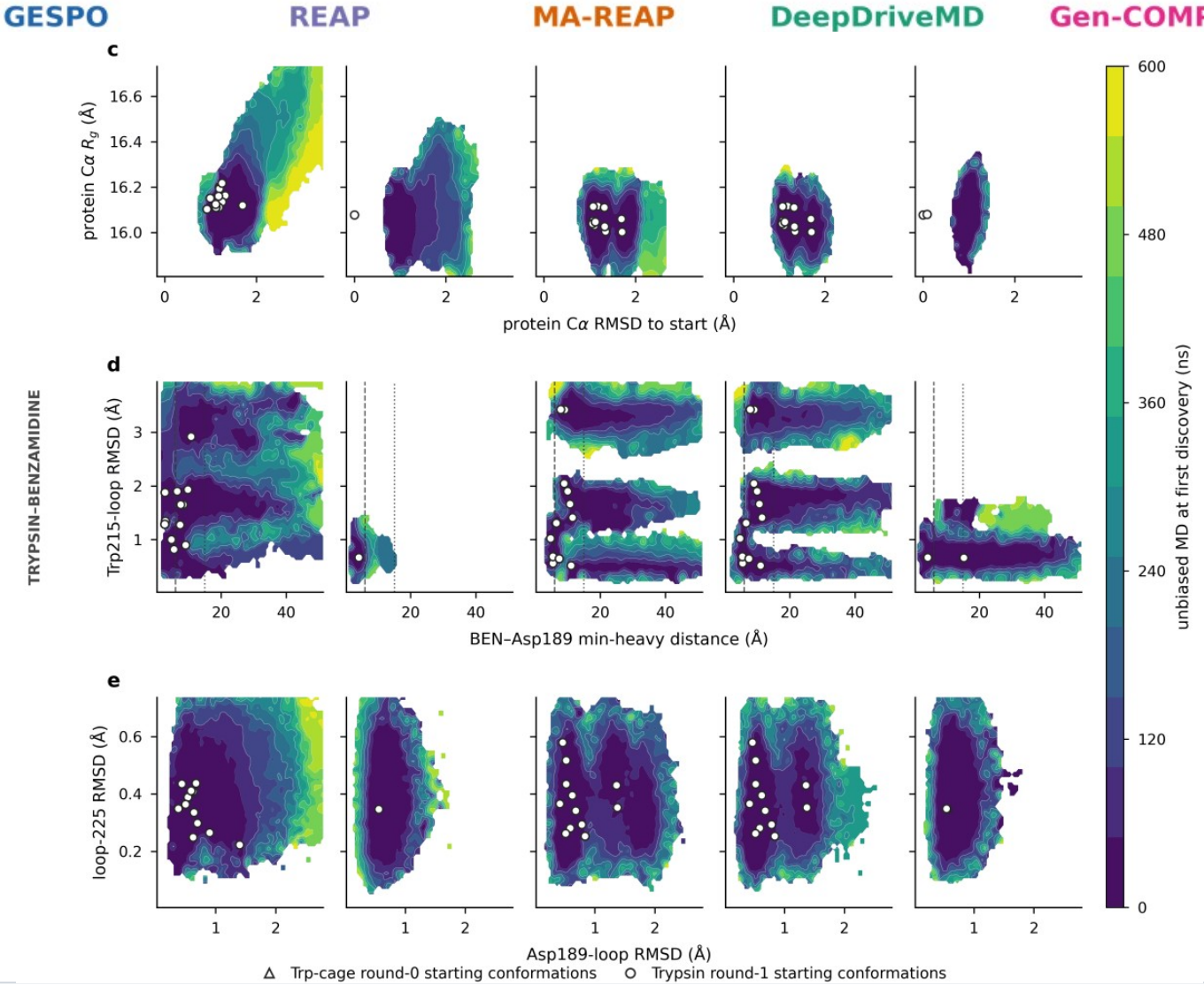
(c) GESPO, holo

Benzamidine shifts sampling further, with both loops displaced together.

Same policy, different system: the apo–holo difference comes from the physics, not the sampler.

(a) 52 unbiased trajectories, 100 µs total (Plattner & Noé, 2015). (b, c) GESPO sampling density $-\ln p_{sample}$, not a free energy. Stars: the seven reference receptor states.

MA-REAP and DeepDriveMD: ligand leaves, gate stays



(d) Ligand vs. gate

MA-REAP and DeepDriveMD carry benzamidine past 40 Å, but their late discoveries sit at nearly fixed Trp215-loop RMSD.

(c) Receptor

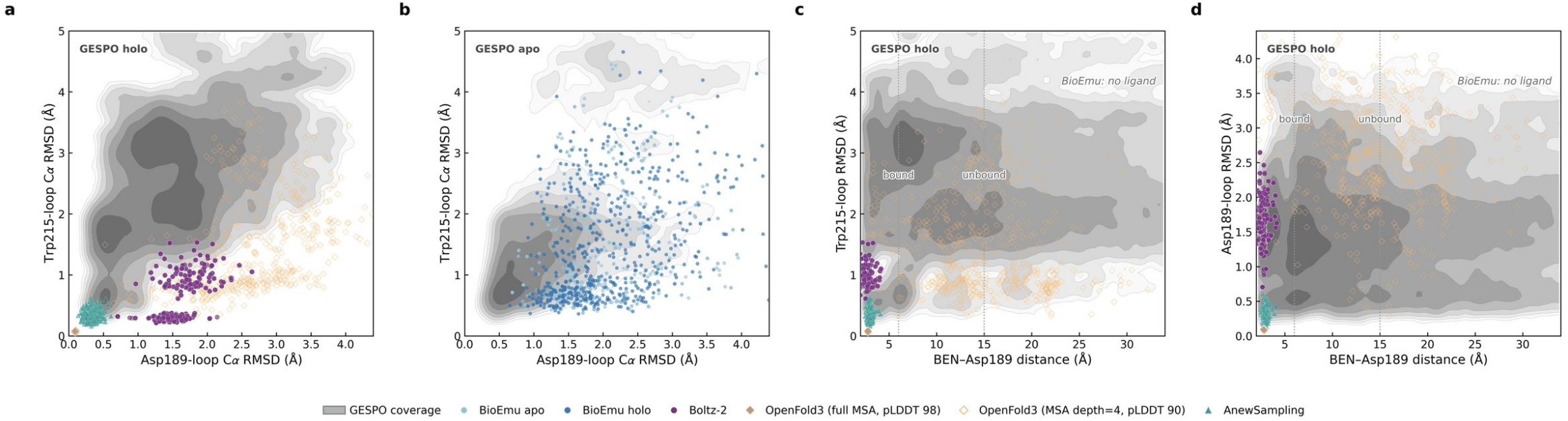
REAP, MA-REAP and DeepDriveMD stay within 3 Å Cα RMSD of the start; GESPO is still moving outward at 10 Å.

(e) Loops

GESPO covers Asp189-loop and loop-225 displacements the baselines do not reach.

Pengmei et al., GESPO (manuscript, 2026); 720 ns per method, matched launch conformations; color: budget at first discovery

Generators supply priors; trajectories supply the mechanism



Co-folding and equilibrium generators

Boltz-2, OpenFold3 (full MSA) and AnewSampling collapse onto the closed, bound pose: benzamidine within ~4 Å of Asp189.

Shallow-MSA OpenFold3

Broad scatter set by an inference setting (MSA depth 4): no S1* region, 44% of samples beyond 15 Å; gone by depth 16.

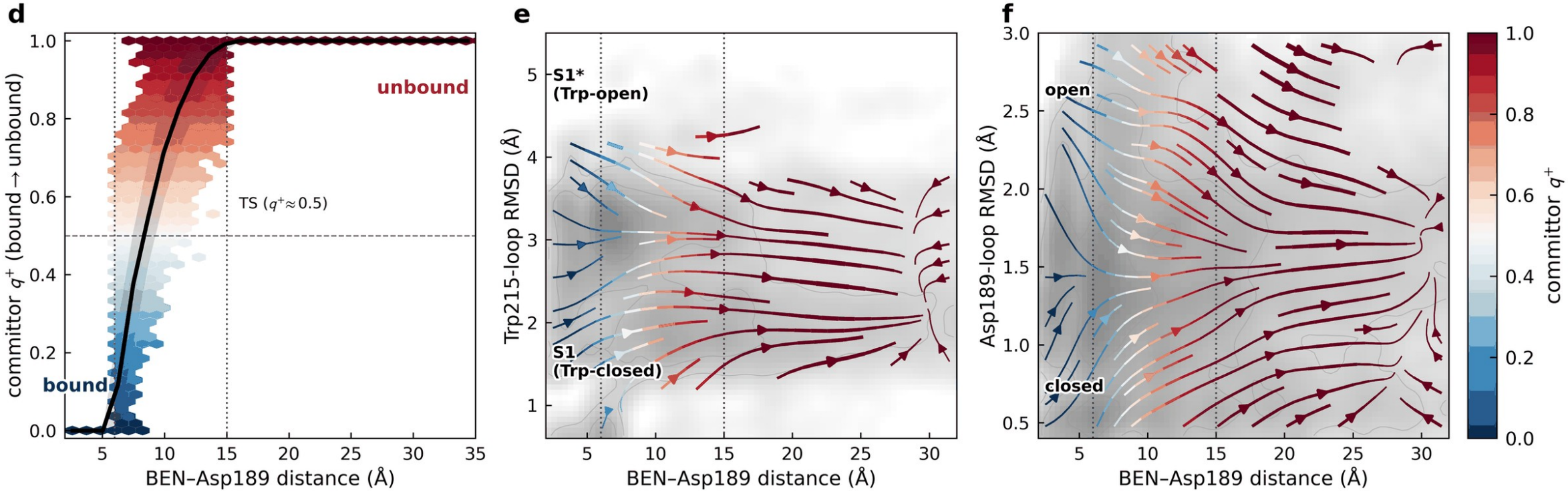
BioEmu

A receptor-only prior: open and closed Trp215 gates without ligand, so it cannot be placed on the association coordinate.

The second binding mode and the route out of the pocket come from the trajectories.

Gray: GESPO coverage. Boltz-2 (Passaro et al., 2025); OpenFold3; BioEmu (Lewis et al., 2025); AnewSampling (2026). Dotted lines: bound (6 Å) and unbound (15 Å) distances.

Ligand distance alone does not determine commitment



(d) Committor vs. distance

q^+ crosses 0.5 near 8–9 Å, as free diffusion between these shells would; the spread at fixed distance stays broad.

(e) Trp215 loop

At equal ligand distance, open-gate configurations sit further toward dissociation.

(f) Asp189 loop

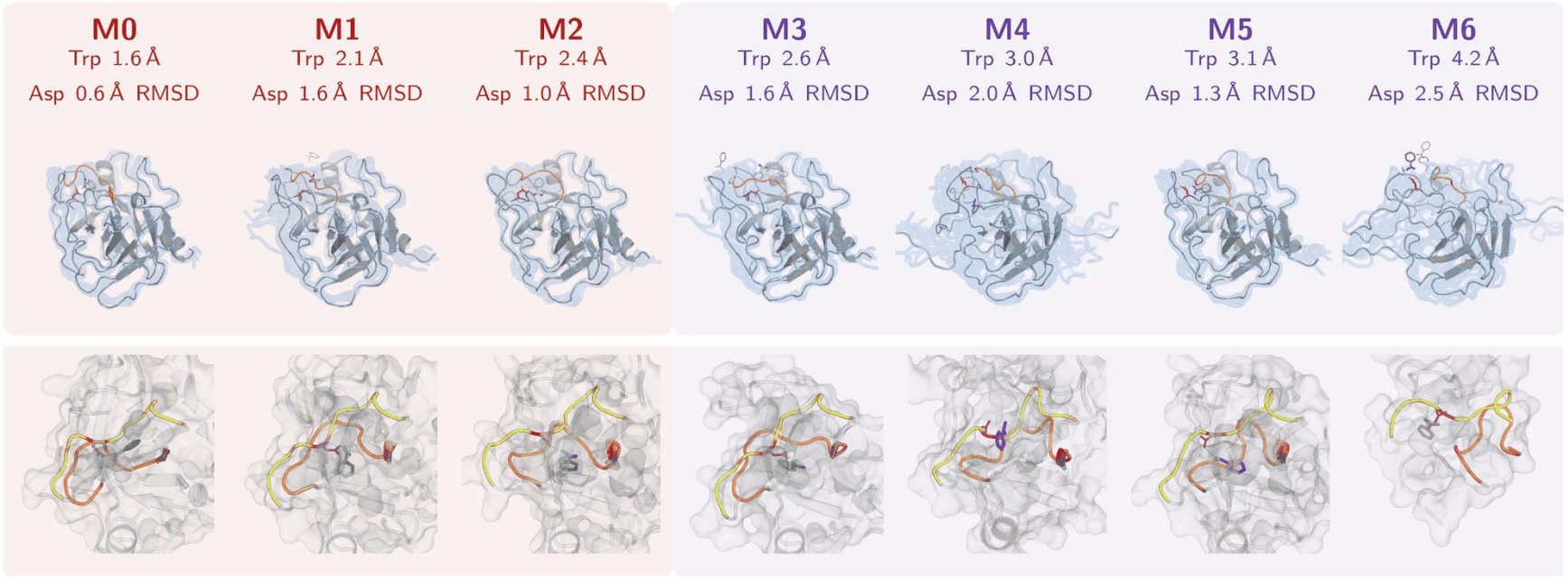
Smaller effect; the Asp189 loop stays near its closed geometry.

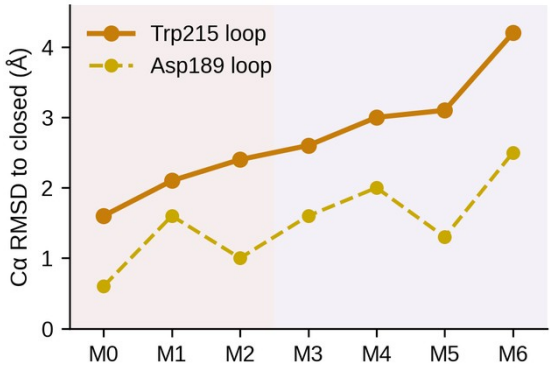
The receptor coordinate carries part of the reaction progress that distance misses.

DGA committor q^+ between bound (d < 6 Å) and unbound (d > 15 Å) sets. (e, f) Mean displacement per lag time, colored by q^+; a drift field, not a probability current. Pengmei et al., GESPO (manuscript, 2026).

The Trp215 loop gates the pocket between S1 and S1*

S1 · gate closed — **S1* · gate open**





Trp215 is the gate

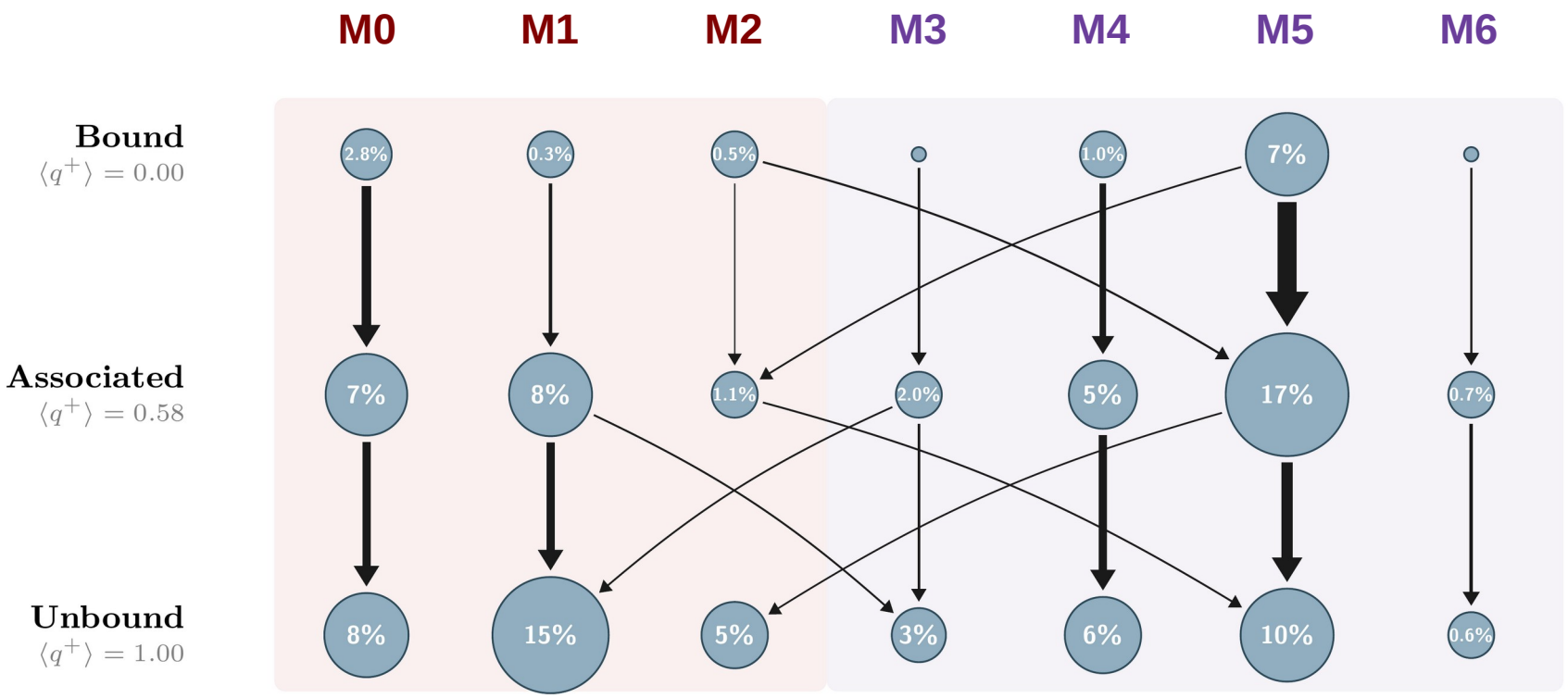
Its RMSD rises monotonically, 1.6 → 4.2 Å. The Asp189 loop stays within 2 Å in all but M6.

Bound benzamidine appears in the closed M0 and throughout the open-gate states: the S1* mode is observed in the trajectories, not inferred from endpoints.

Seven receptor states, ordered by how far the Trp215 gate opens.

SPIB on Geom2Vec features (holo), seven macrostates as in Plattner & Noé (2015). Pocket: Asp189 loop yellow, Trp215 loop orange. Pengmei et al., GESPO (2026).

Most dissociation flux leaves the pocket through M5



Reading the network

Rows: committor tiers, bound → associated → unbound. Arrow width: net reactive flux; circle area: stationary weight.

Result

The largest flux passes through the open-gate state M5; M0 and M4 follow; M2 carries the least. All arrows point toward dissociation.

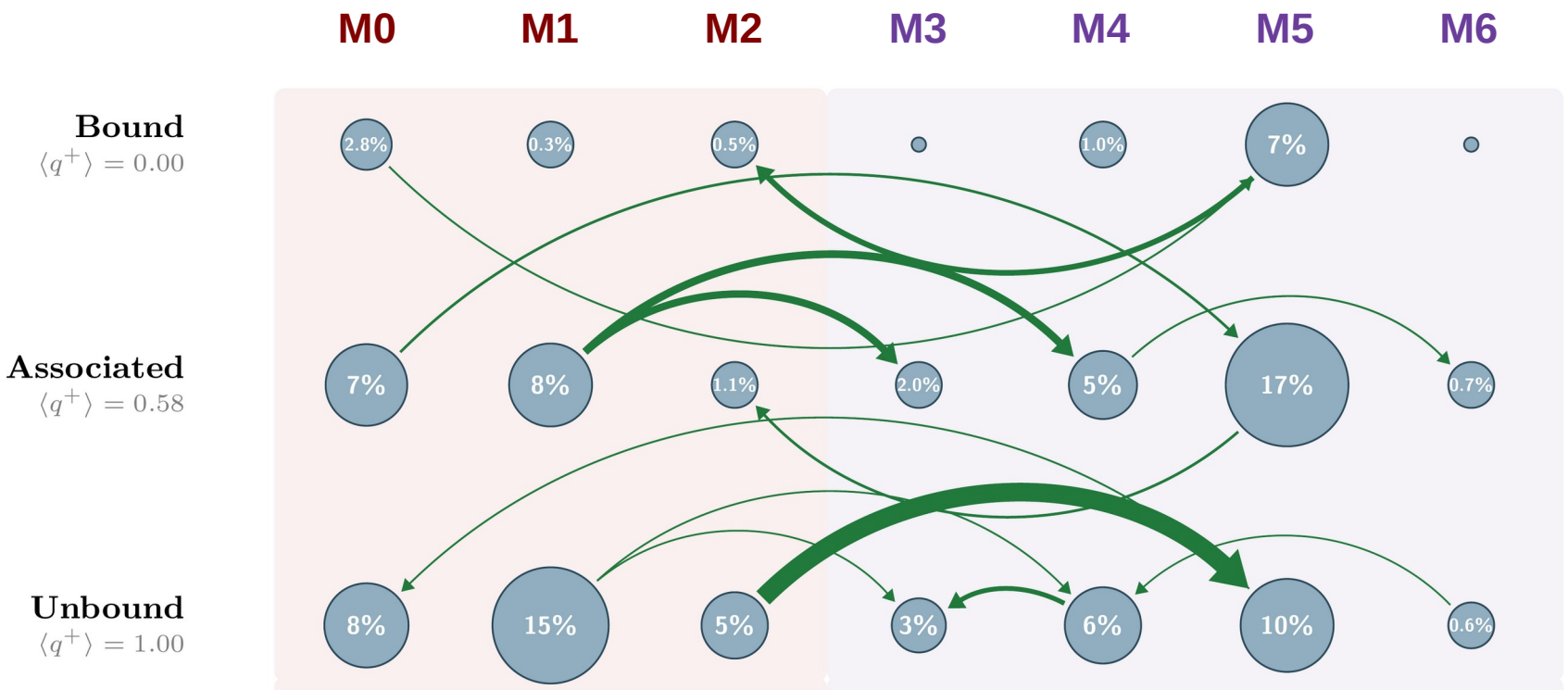
Scope

Weights are reweighted from adaptive data (π/p_{sample})

Dissociation runs mainly through the open-gate state M5; M0 and M4 are secondary routes.

Net reactive flux $\max(f_{ij} - f_{ji}, 0)$, $f_{ij} = \pi_i q^-_i T_{ij} q^+_j$, after reweighting each frame by π/p_{sample}. Pengmei et al., GESPO (manuscript, 2026).

The receptor interconverts across the full S1–S1* range



Reading the network

Green arrows: net conformational exchange between macrostates. Width: size of the net current.

Result

Every macrostate appears in the associated tier; exchange is strongest between M2 and M5.

The gate opens and closes throughout binding

Same macrostates, tiers and reweighting as the reactive-flux network. Pengmei et al., GESPO (manuscript, 2026).

What the evidence supports, and what it does not yet

Limitations

1. **Statistical scope**
   Pooled frames from adaptive sampling are not equilibrium samples; populations, free energies and rates need reweighting or independent validation.

2. **Force field fidelity**
   The quality of potential function limits the correct distribution that we are sampling from.

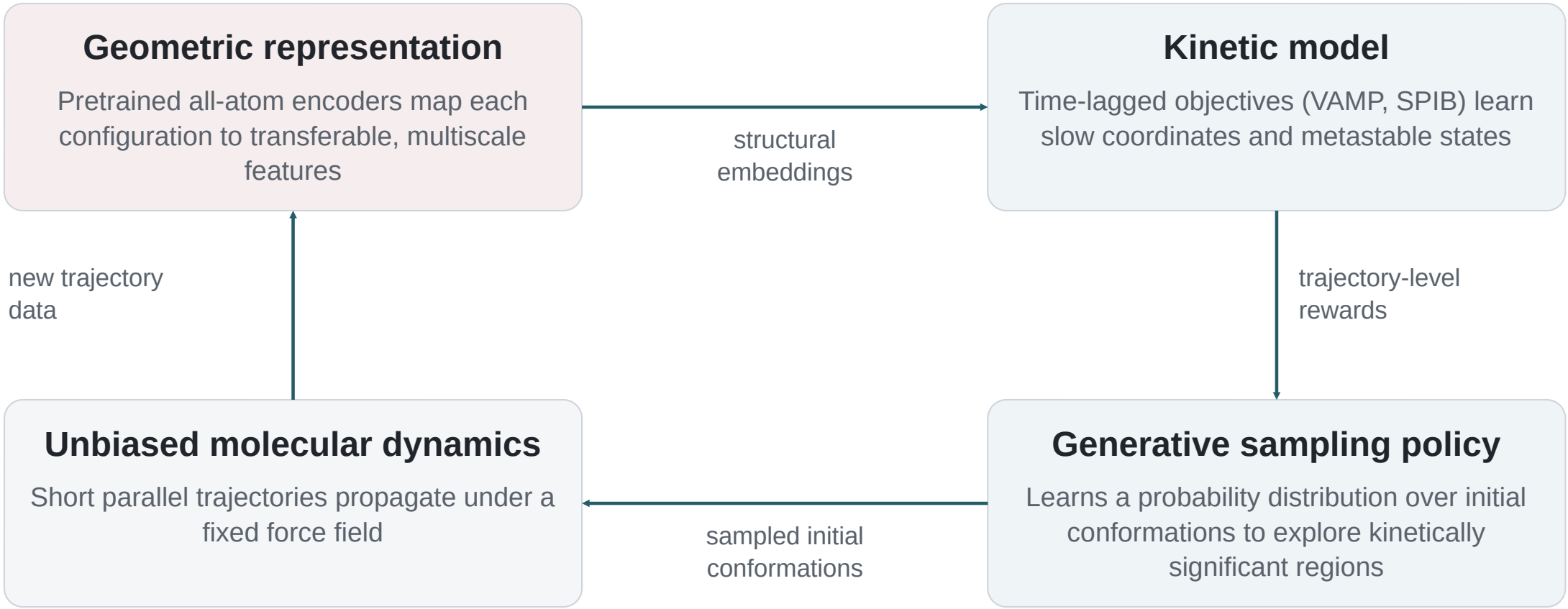
Future directions

1. **Estimators for adaptive data**
   Connect adaptive sampling to statistically justified equilibrium and kinetic estimators.

2. **Richer actions**
   Let the policy also choose resolution, number of trajectories and duration.

3. **Generators with kinetic feedback**
   Post-train structure predictors toward initial conformations that are informative under the dynamics.

Representation and reinforcement learning meet at kinetics



Features retain slow motion; kinetic evidence decides where to sample.

Thank you

Aaron R. Dinner

Advisor

Committee

Suriyanarayanan Vaikuntanathan
Rebecca Willett

Geom2Vec and token merging

Chatipat Lorpaiboon · Spencer C. Guo · Jonathan Weare

GESPO

Nhan D. Nguyen · Zhengyuan Shen · Jonathan Weare

All-atom pretraining

Zhengyuan Shen · Zichen Wang · Marcus Collins · Huzefa Rangwala

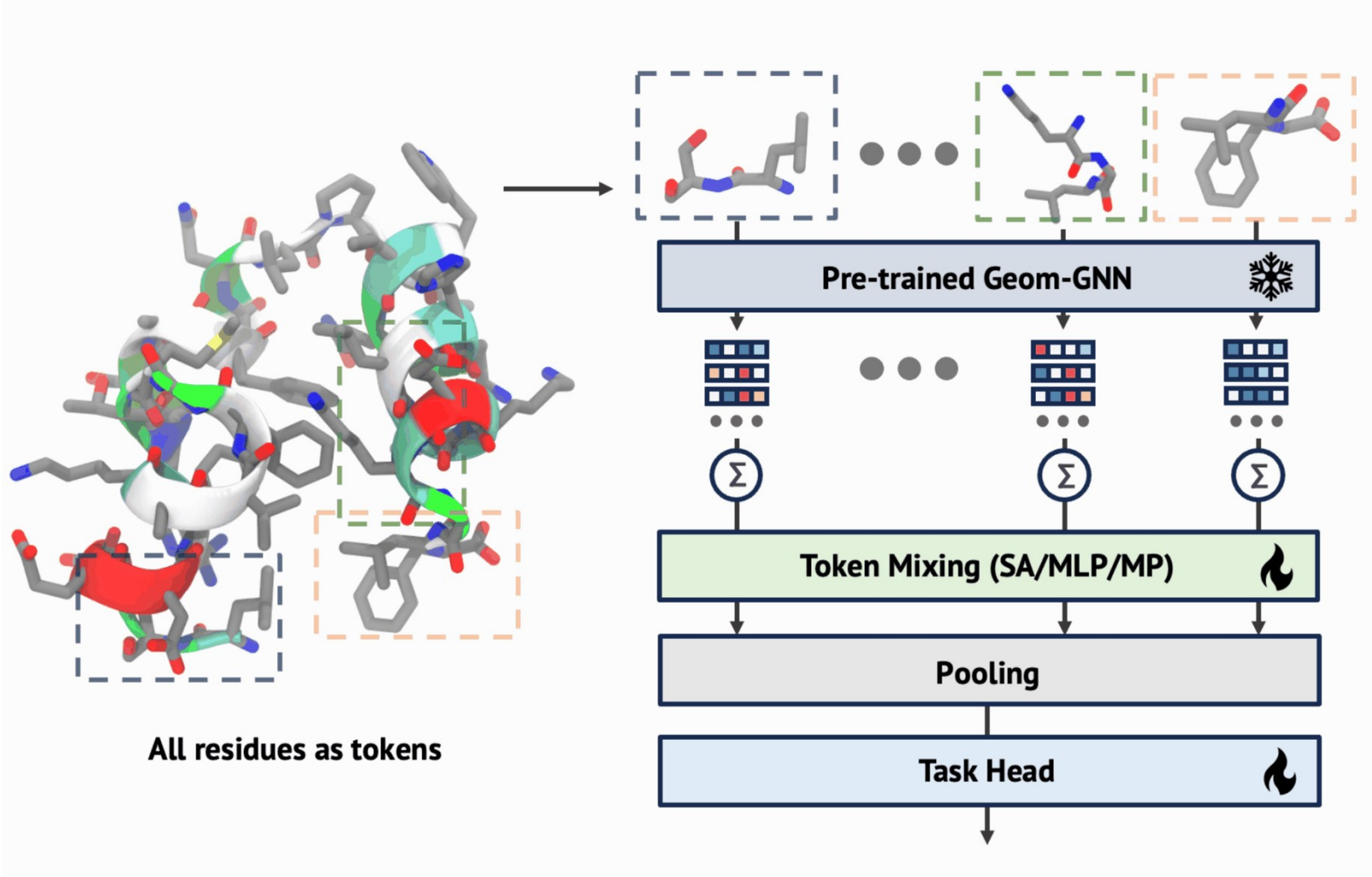
Data and computing

D. E. Shaw Research (trajectories) · UChicago RCC (Beagle-3) · AWS

Questions

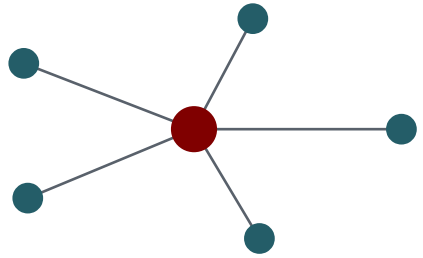
Funding: NIH R35 GM136381; NSF DMS-2054306; UChicago Data Science Institute AI+Science Research Initiative; AWS.

Frozen residue tokens feed a trainable token mixer



Local message passing has distinct information limits

Local update



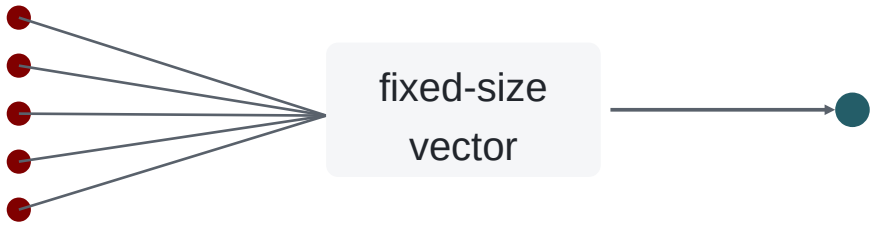
Aggregate neighboring features.

Under-reaching



Distant domains require more hops.

Over-squashing



Many signals share a fixed-size channel.

Over-smoothing



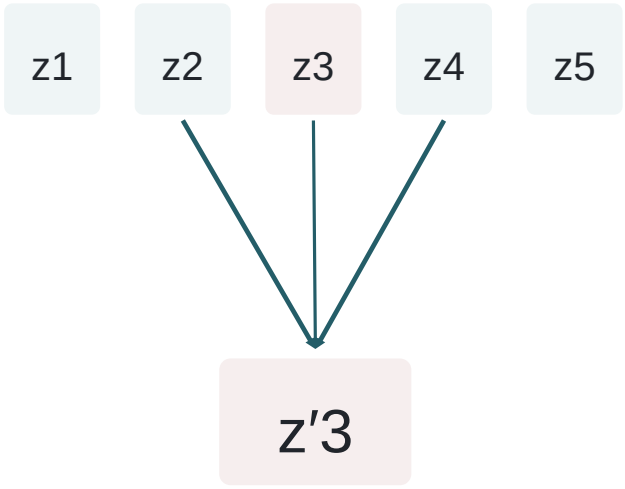
Repeated mixing erases feature contrast.

Dissertation Ch. 1, Fig. 1.2; over-squashing: Alon & Yahav (2021); over-smoothing: Oono & Suzuki (2020).

Sequence, graph and attention updates mix different inputs

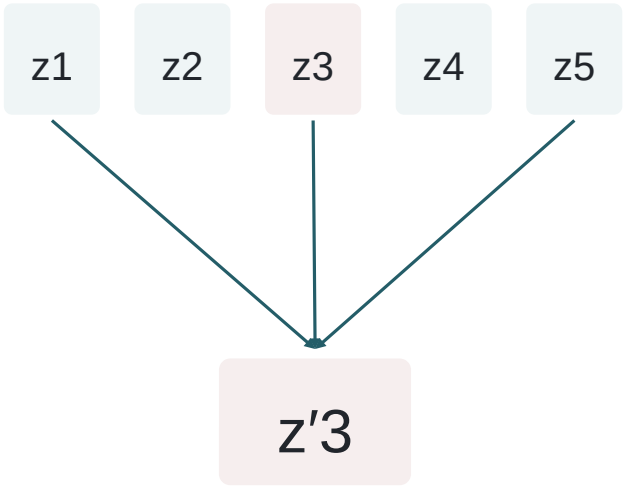
Sequence-local

Nearby sequence positions



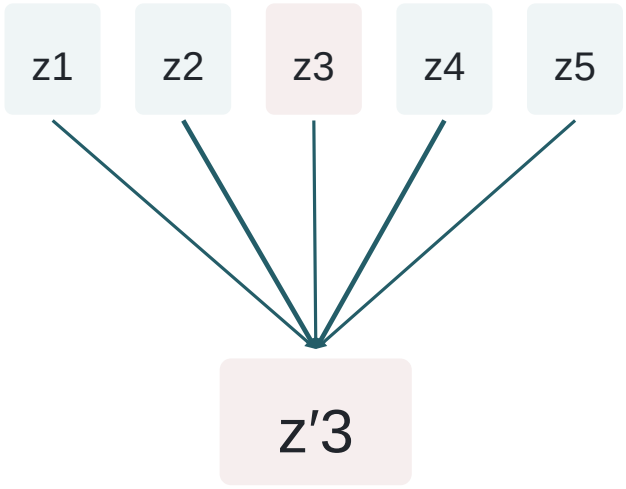
Graph

Bond or contact neighbors



Self-attention

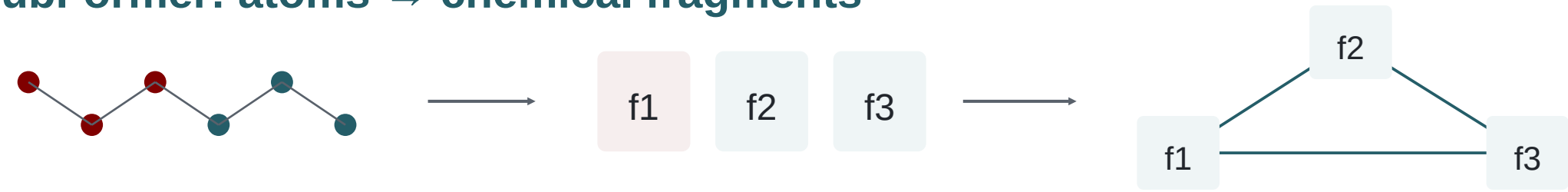
All unmasked tokens



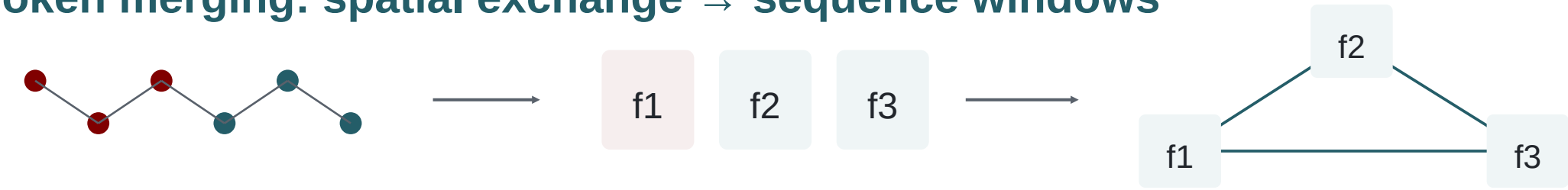
Dissertation Ch. 1, Fig. 1.3 (a–c).

Hierarchy changes the representation, not the MD resolution

SubFormer: atoms → chemical fragments



Token merging: spatial exchange → sequence windows



local atomic / residue information

learned pooled representation

global mixing

Dissertation Ch. 1, Fig. 1.3 (d, e). SubFormer: Pengmei et al. (2023); token merging: Pengmei et al., J. Chem. Phys. (2026).

The kinetic score uses centered empirical covariances

$$u = \chi(X_t), \qquad v = \chi(X_{t+\tau})$$

$$C_{00} = \langle (u - \mu_u)(u - \mu_u)^\top \rangle$$

$$C_{0\tau} = \langle (u - \mu_u)(v - \mu_v)^\top \rangle$$

$$C_{\tau\tau} = \langle (v - \mu_v)(v - \mu_v)^\top \rangle$$

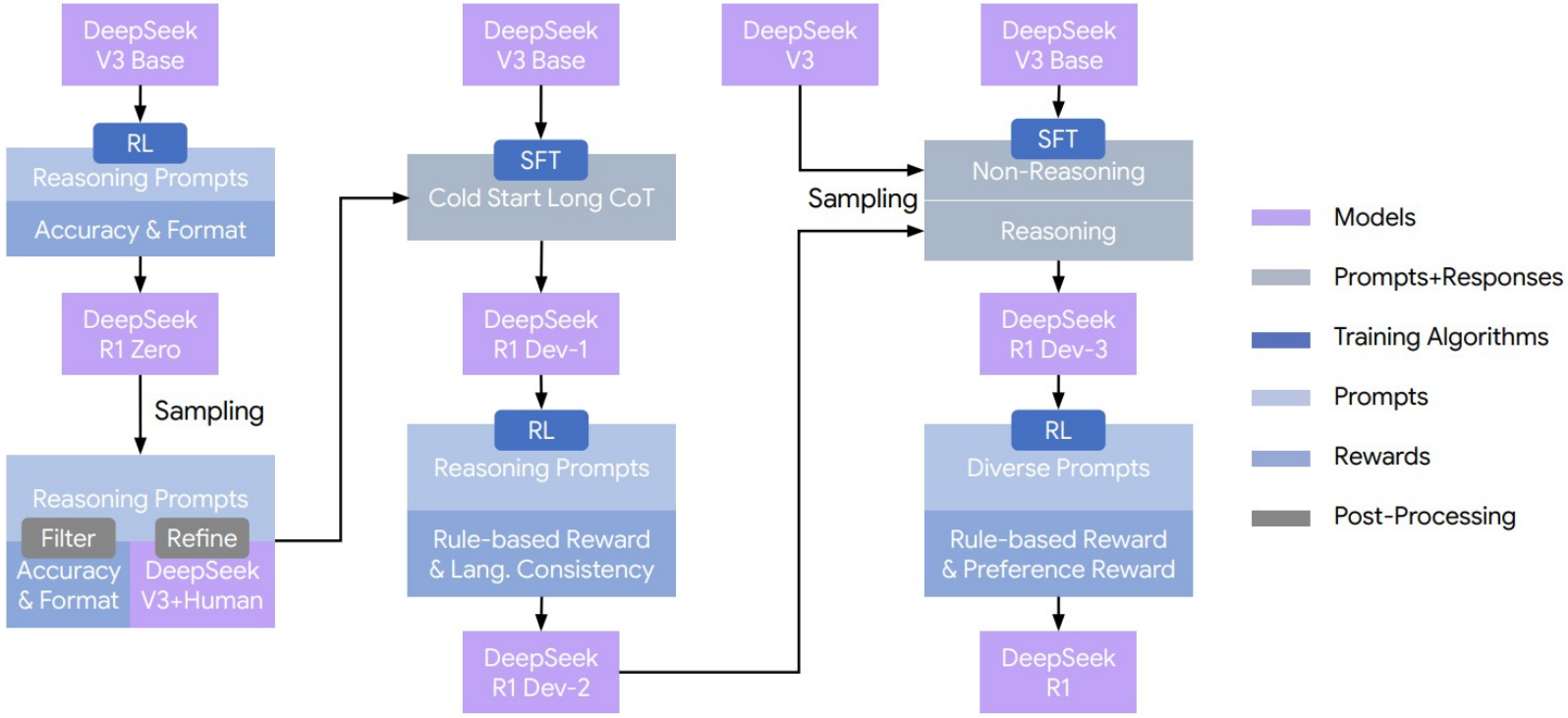
Removing a segment changes the empirical means as well as the cross-products.

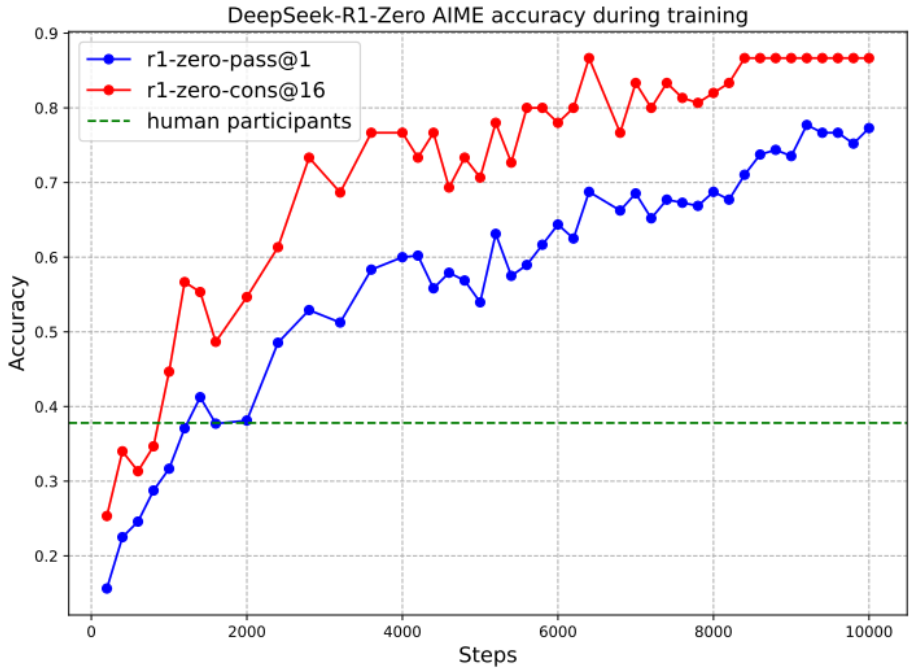
Pengmei et al., GESPO (manuscript, 2026), Eqs. for the VAMP-2 estimator.

GESPO settings for the two production campaigns

	Trp-cage	Trypsin–benzamidine
Force field, T	CHARMM36, TIP3P; 300 K	AMBER14 + GAFF2, TIP3P; 298 K
Trajectories	G = 12 per epoch, 200 ps each	G = 12 per epoch, 2 ns each
Campaign	3 rounds (10, 12, 24 epochs); 115.2 ns	2 rounds × 15 epochs; 720 ns
Initial data	4.8 ns bootstrap (discrete-state variant)	100 ns well-tempered metadynamics (excluded from kinetics)
Policy	4 layers, d = 256, 8 heads, 8 mixtures; 3 descriptors + 20 residues	5 descriptors + 223 residues + 2 ligand torsions
GRPO	lr 2×10^{-4}; 10 iterations/epoch; ε = 0.1; KL 0.2; entropy 0.1–0.2	lr 5×10^{-5}; 20 iterations/epoch; ε = 0.1; KL 0.1; entropy 0.15
Realization	Restrain bank frame toward proposal, release before production	Same; ligand–pocket flat-bottom restraint only during realization

Analogy: RL post-training of a language model (DeepSeek-R1)

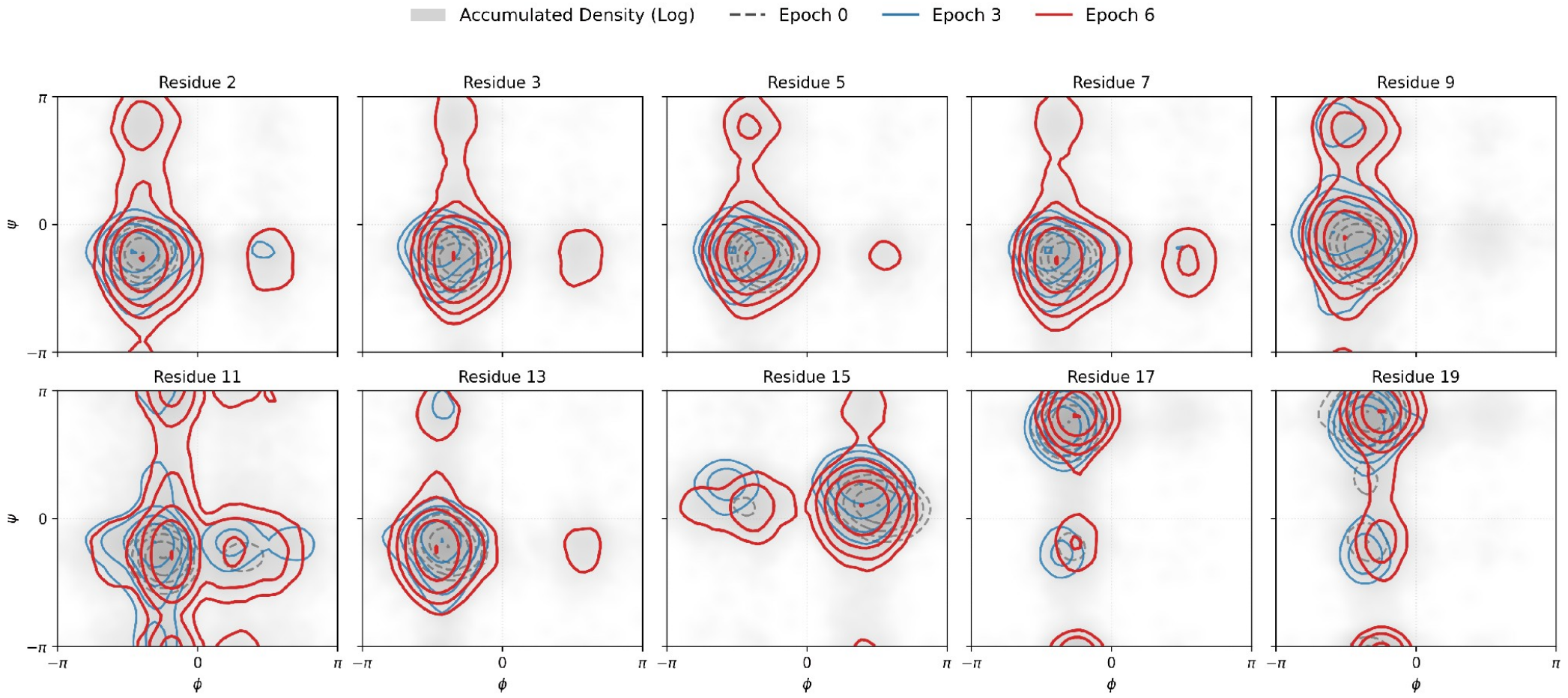




GESPO borrows the update rule (GRPO), not the model or the data.

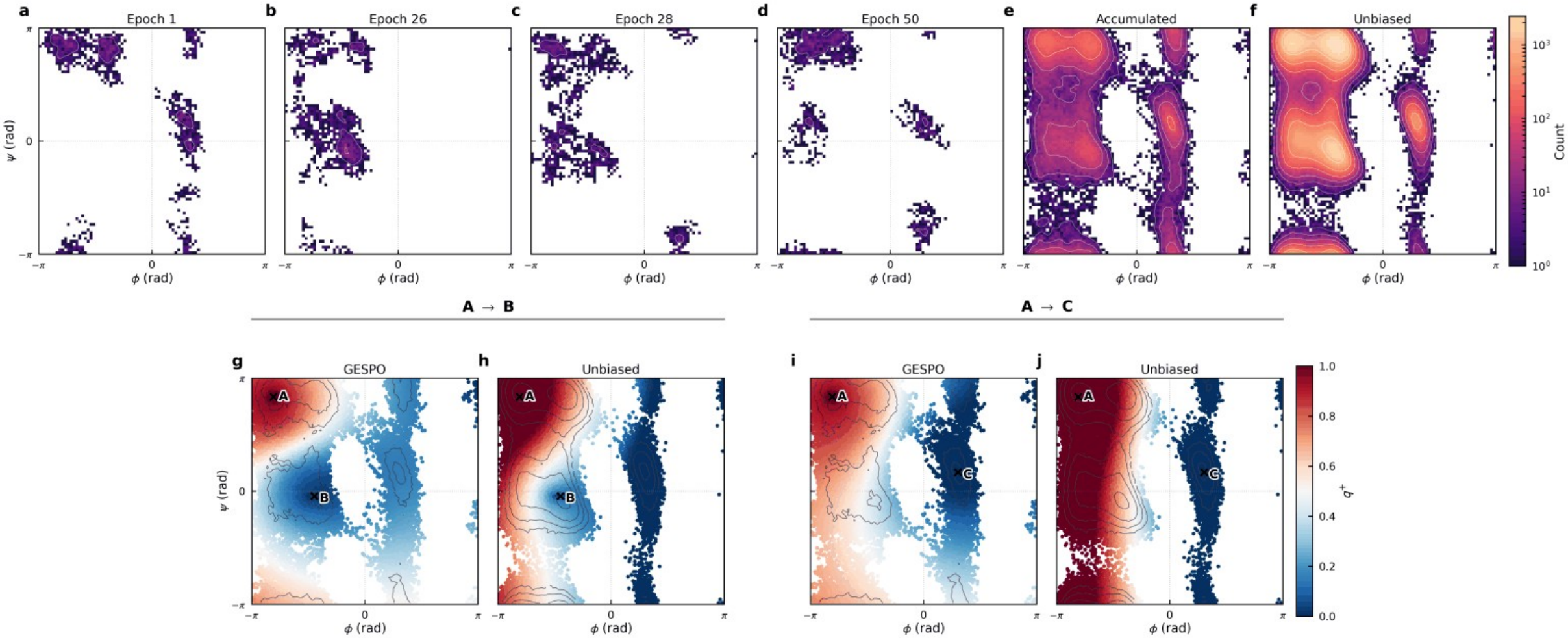
Figure: DeepSeek-AI, DeepSeek-R1 (2025); shown as an analogy only

Launch-policy torsion marginals for ten Trp-cage residues

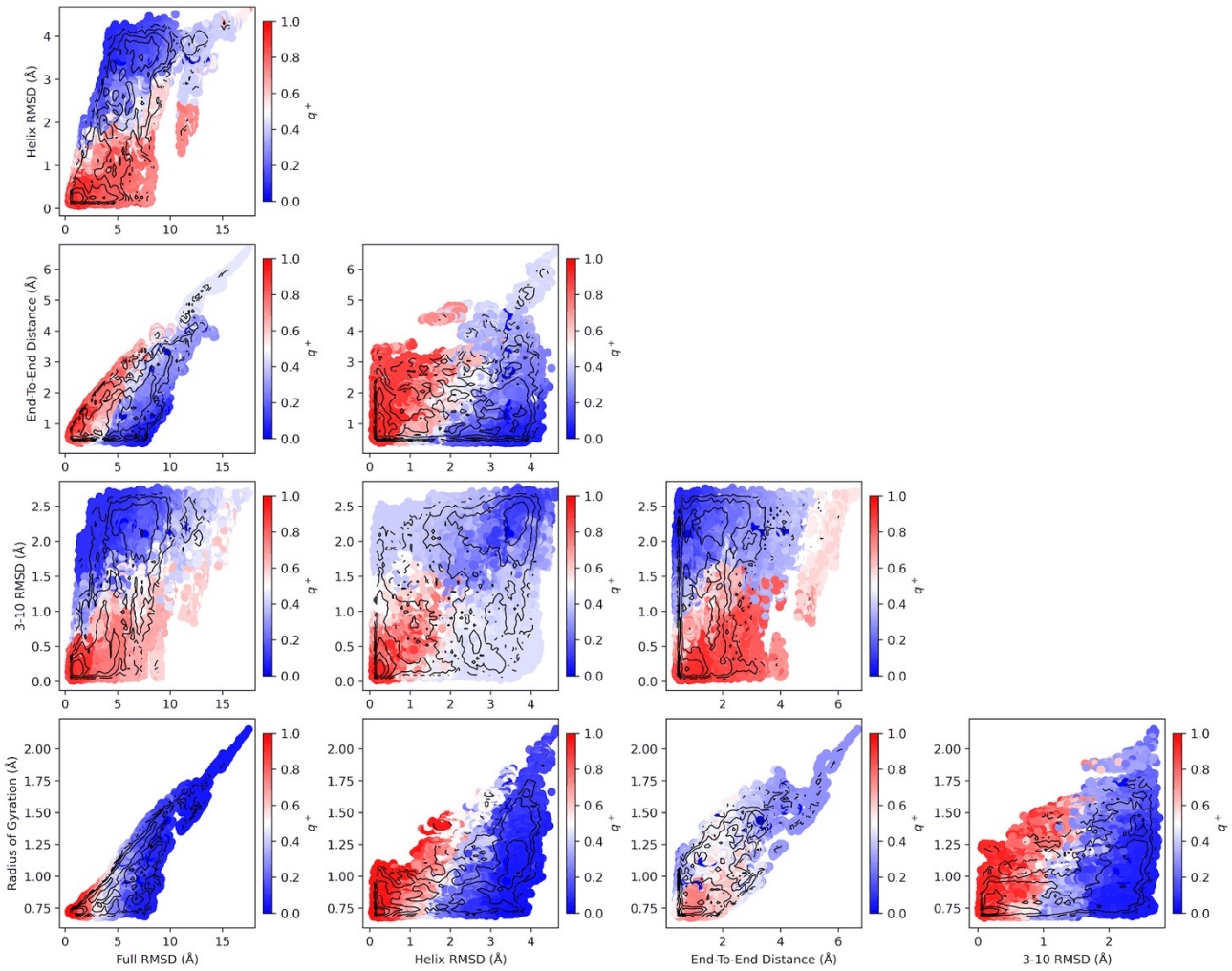


Pengmei et al., GESPO (manuscript, 2026). Epochs 0 (dashed gray), 3 (blue) and 6 (red) over the accumulated sampling density.

Alanine dipeptide: basins and committors from 0.8 ns



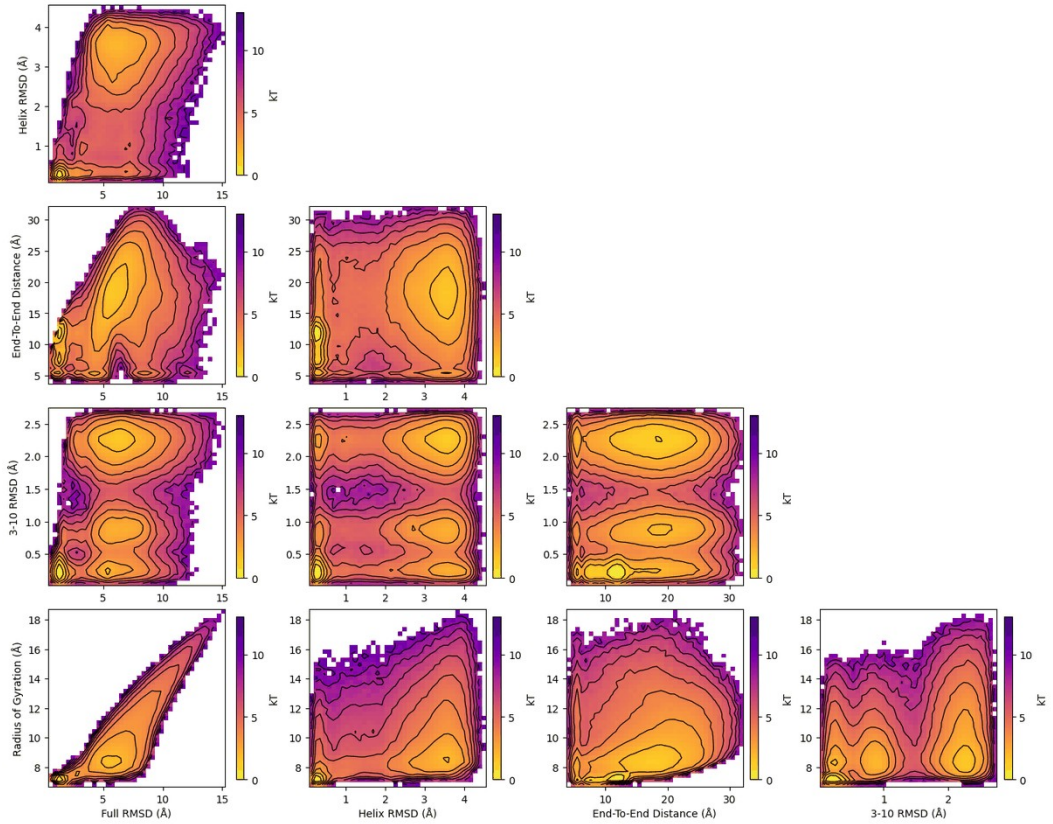
The Trp-cage committor varies smoothly across coordinates



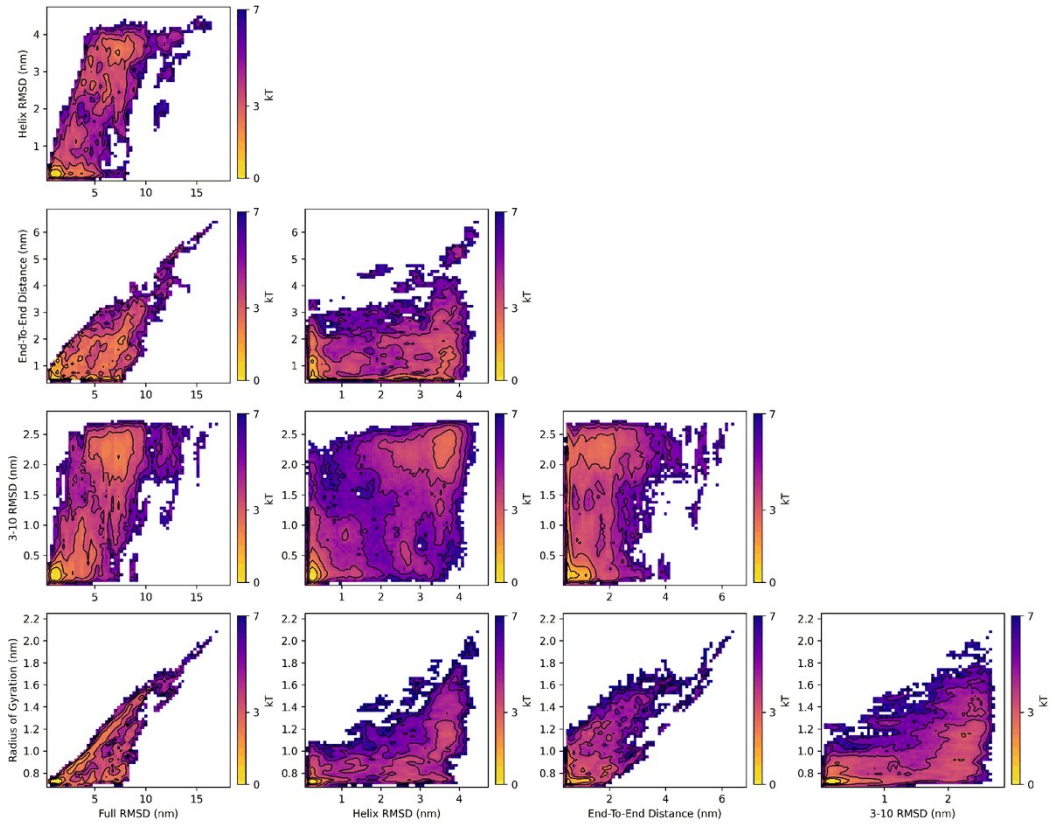
Pengmei et al., GESPO (manuscript, 2026). DGA committor q^+ on pairwise projections of the GESPO data.

The 208 µs reference and GESPO share one topography

Anton reference: 208.8 µs

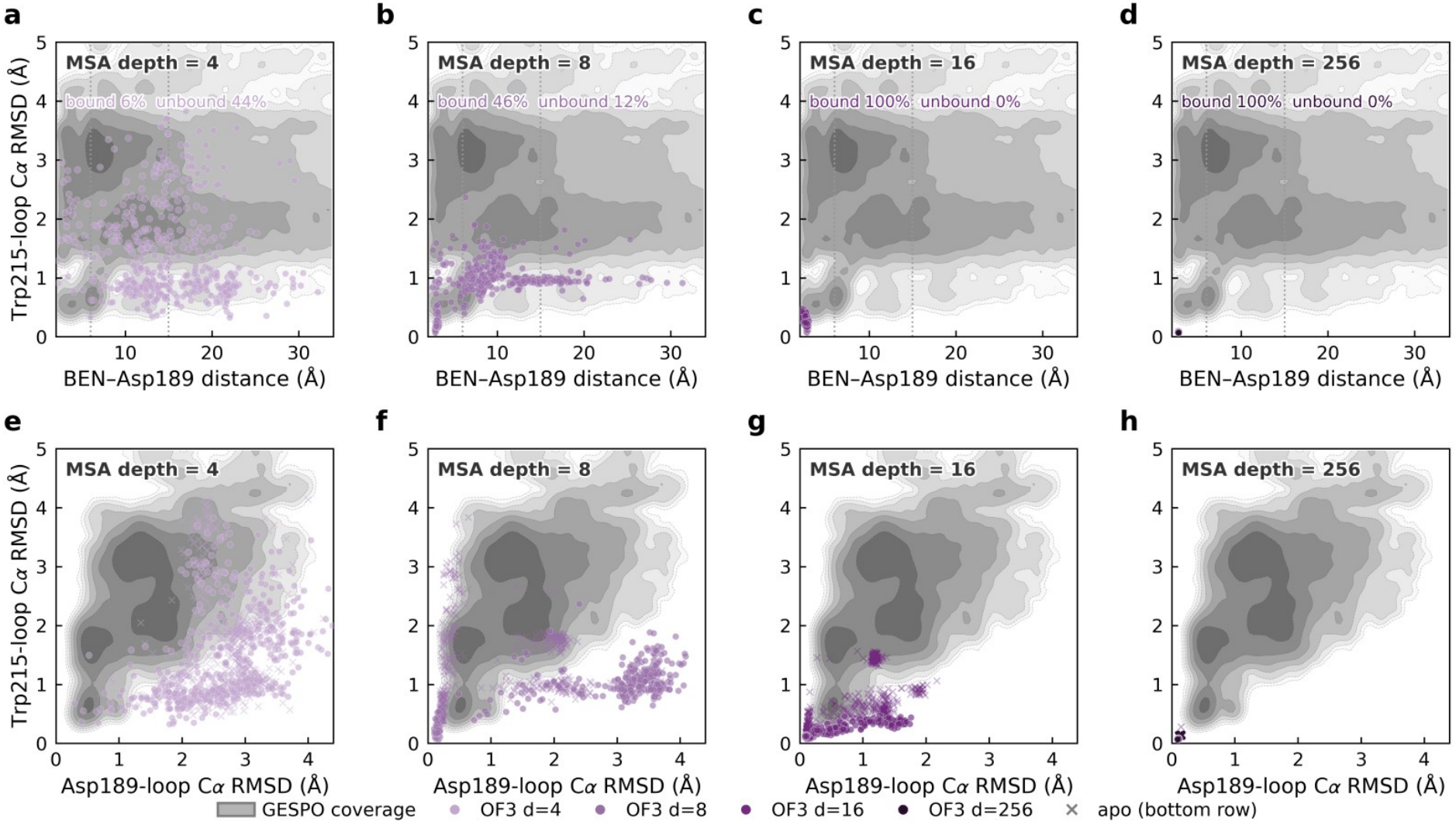


GESPO: 115.2 ns of short trajectories



Pengmei et al., GESPO (manuscript, 2026). Left: Anton, K8A TC10b, CHARMM22*, 290 K (Lindorff-Larsen et al., 2011). Right: GESPO, CHARMM36, 300 K; sampling density, not equilibrium.

OpenFold3 sample spread is set by the MSA depth



Pengmei et al., GESPO (manuscript, 2026). Top: holo, benzamidine–Asp189 distance; bottom: loop RMSDs. Gray: GESPO coverage.